Art of the Possible Handbook

AFSCH60-101
1 OCTOBER 2026

USAF AoP
The AFSC Way

Air Force Sustainment Center

To the Men and Women of the Air Force Sustainment Center,



As Airmen in AFSC, flying and fixing is at the very core of what we do to ensure our Air Force, *the world's greatest Air Force*, is ready to fight. From projecting power from our installations, modifying aircraft in our production lines, supplying serviceable spare parts, to providing more capable software, every part of AFSC impacts our Air Force's ability to deliver combat power.

Art of the Possible (AoP) is not what we do; it is how we do everything! Our mission execution, centered on AoP, is built on high standards – how we perform our daily duties, how we ready ourselves, and how we treat each other.

AoP brings standard work to complex processes, and enables us to continuously improve, break through constraints, and identify ways we can do things better, safer, and faster. The AoP Leadership Model shapes our culture and guides how we interact with each other and those we support. The values of teamwork, accountability, respect, transparency, credibility, engagement serve as guiding principles for all we do.

Air Force readiness starts with our readiness. Our readiness starts with successful implementation of AoP methodologies across all operations. AoP is the fundamental basis of how we operate across AFSC. It requires leadership and commitment at all levels. It is both a philosophy and a methodology that enables us to achieve exceptional results for our Air Force. I expect every Airman in AFSC to internalize AoP and practice its principles every day.

Mission and environments continue to change, and so must we. We must build on current efforts to ensure AFSC is ready and "running to the sounds of the guns" as our Air Force and our nation face the ever-changing landscape of the globe. We will also look to the future to enhance work environments, tools, and support processes to strengthen our own readiness.

This handbook is the cornerstone of institutionalizing AoP across our enterprise. It is the "playbook" that allows us to operate as one team. It reflects collective experiences, lessons learned, best practices, and important content on how AoP will be applied in every work area. Our customers and our suppliers benefit when we speak with a consistent voice, use a standard set of operating principles, and constantly look to enhance processes. **When you need a "north star" on how to operate or encounter a challenge in your mission area, go to this handbook first for the solution – it's in here!**

Through our commitment to disciplined implementation, we will continue to operationalize AoP and drive continuous improvement and readiness focus into our mission every day. Readiness begins with us. **Deliver Airpower … stay ready!**

JENNIFER HAMMERSTEDT
Lieutenant General, USAF
Commander

BY ORDER OF THE COMMANDER
AIR FORCE SUSTAINMENT CENTER

AIR FORCE SUSTAINMENT CENTER
HANDBOOK 60-101

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Standardization

ART OF THE POSSIBLE

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This is a newly revised handbook and should be reviewed in its entirety. As stated in Joint Publication (JP) 4-0, Joint Logistics, "the relative combat power that military forces can generate against a threat is constrained by their capability to plan for, gain access to, and deliver forces, materiel and commercial capabilities to points of need." Art of the Possible (AoP) is the constraints-based management system used by the Air Force Sustainment Center (AFSC) to provide effective support to the delivery of combat power by the warfighter. This handbook implements Department of the Air Force Policy Directive (DAFPD) 60-1, Air Force Standardization Program and prescribes minimum requirements for implementing AoP, a standard constraints-based management system for managing, conducting, tracking, and reporting workload performed within AFSC. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) listed above using the AF Form 847, Recommendation for Change of Publication. Route AF Form 847s from the field through the appropriate chain of command. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Instruction (AFI) 33-322, Records Management and Information Governance Program, and disposed of In Accordance With (IAW) Air Force Records Disposition Schedule (RDS) located in the Air Force Records Information Management System (AFRIMS).

SUMMARY OF CHANGES

This document has been entirely rewritten to provide a more comprehensive and structured "playbook" for implementing the Art of the Possible (AoP) methodology. It serves as the keystone concepts for AFSC's constraints-based management system. It should be reviewed in its entirety.

1. Scope. This publication is the keystone document for AFSC's AoP. It provides overarching concepts on constraints-based management of most critical processes. It provides the foundation, fundamentals, and core tenets that guide commanders and directors in implementing, executing, and maturing AoP.

2. Purpose. This publication has been prepared under the direction of the commander of the AFSC. It sets forth AFSC concepts for the performance of constraints-based management and provides the basis for the implementation, execution, and maturation of AoP within AFSC units. It provides guidance for the management of AFSC most critical processes. It provides the framework within which critical processes can be optimized to support Air Force operations throughout the world. This publication is intended to provide guidance to AFSC commanders, directors, and their staff for constraints-based management of most critical processes.

3. Application.

3.1. This handbook provides Air Force Sustainment Center (AFSC) guidance, instructional information, and procedural methodologies regarding the Art of the Possible (AoP) constraints-based management system. It serves as an operational reference to assist personnel across work centers in managing, conducting, tracking, and reporting workload performed within AFSC organizations.

3.2. The use of the AoP Handbook is directed by Air Force Sustainment Center Instruction (AFSCI) 60-101, *Art of the Possible,* and should be used by AFSC senior leaders to create a culture that relies on the skills, abilities, and forward thinking of the entire enterprise to create the teamwork necessary to enable AoP. If conflicts arise between the contents of this publication and the contents of complex, wing, or directorate publications, this publication should take precedence unless the commander of the AFSC has provided more current and specific guidance.

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Executive Summary

Art of the Possible (AoP) is a leadership and performance management system designed to align organizations around achieving breakthrough results by focusing on what truly drives success. AoP challenges leaders to shift from a mindset of working harder to one of working smarter—directing energy toward the constraints that unlock the greatest gains for the mission.

At its heart, AoP depends on leaders who take ownership, model accountability, and foster cultures where teams embrace improvement and align their efforts with mission priorities. This approach transforms confusion into clarity, frustration into flow, and scattered activity into purposeful outcomes.

AoP recognizes every system has a constraint, and performance of the whole organization is governed by how that constraint is managed. By applying disciplined principles such as controlling Work In-Process (WIP), releasing only supportable work, assigning work using a synchronized plan, and resolving issues quickly, leaders ensure scarce capacity is focused on what matters most.

AoP leverages Continuous Process Improvement (CPI) as a way of thinking and working. Through CPI, leaders and teams diagnose root causes, eliminate waste, and increase long-term capacity, turning constraint management into a proactive cycle of learning and sustainable progress.

Ultimately, AoP is about demonstrating that extraordinary results can be achieved when leaders focus on constraints, discipline, and culture. By doing so, organizations not only meet today's demands but also create the conditions for tomorrow's success.

Chapter 1

INTRODUCTION

1.1. What is Art of the Possible?

1.1.1. For decades the Air Force, as well as industries across the globe, have been in a continuous search for better ways to do business. There has been a constant stream of new concepts that have consumed the time and attention of the industrial world. Each passing year seemed to bring the next big thing that promised to transform the way business was done. Concepts such as Total Quality Management, Lean, ToC, 6 Sigma, as well as dozens of variations and combinations of these, have been at the forefront of every organization in the Air Force. Each of these brought great promises but equally large challenges. Every new initiative brought a new vocabulary and new tools that must be learned. With them also came a growing mountain of resistance to the next "flavor of the month."

1.1.2. When AFSC stood up in July 2012, it was a collection of individual depots and supply chain pieces operating with individualized ideas that had been developed through 30 years of innovation initiatives. This merging of supply chain functions under one center presented the opportunity to standardize business practices and adopt operating principles that would become the basis of a new culture: the Art of the Possible. AoP is not a new initiative, but a universal management system that establishes a standardized set of operating principles for AFSC. It takes the gold standard concepts that have been learned through years of experience and combines them into one language and customizes them to the work done in AFSC. AoP is a universal set of best practice standards that form the AFSC management system. In accordance with AFSCI60-101 *Art of the Possible*, AoP should be used to govern all AFSC processes.

1.1.3. AoP is a constraints-based management system where achieving increased throughput is the primary focus. AoP is based on the concept of gaining an accurate understanding of the flow of work and then locating the constraint that interrupts the flow and prevents effective execution of the mission. Successful mission execution is defined by achieving the common goals of the organization to include speed, safety, quality, and cost effectiveness. AoP is the persistent challenging and assessing processes so constraints can be identified. Once the constraint is identified, specific focusing steps are used to resolve the constraint with a continuous eye on the goals of the organization.

1.2. Why AoP?

1.2.1. Commitment to AoP stops the churn caused by the continuous introduction of new concepts and buzz words. It creates a common language and a set of guiding principles that are backed by science to direct the approach to work and improvement of work. AoP is a methodical approach to our business; a science behind our operations based on sound "flow" principles utilizing a constraints-based management philosophy that leads to a predictable output. Daily identification and elimination of process constraints with a focus on Continuous Process Improvement (CPI) is essential for success. AoP allows the user to simplify complex processes into visible and understandable content that can be monitored and communicated with others.

1.2.2. AoP accomplishes its objectives by increasing throughput. One way to increase throughput is by increasing speed. But a word of caution, speed is not about working faster, cutting corners, or taking risks. The term speed in the AoP lexicon is meant to be synonymous with efficient processes that promote throughput paced to a Road to… Goal. In its most basic sense, speed equals reduced flowtime. AoP creates a methodology that measures performance in a manner that focuses the organization on the weakest link in their processes. This focus leads to process improvement initiatives that affect the speed of throughput for the organization. To achieve speed, issues that affect the critical path of the process must be resolved quickly to allow the product to continue moving forward unhindered.

1.2.3. Although AoP is a management system based on scientific principles, it is nothing without engaged leadership. AoP includes developing effective leadership skills. Leadership is the fuel that drives the organization. Leadership must be engaged in understanding the health of the system and relentlessly striving to make it better through constraint resolution. Leaders must be proactive in working within and across organizational boundaries. All stakeholders must understand the goals and be engaged in meeting those goals. It is essential that all leaders at all levels share this responsibility. Chapter 2 will provide details of what engaged leaders should be and what they should do.

1.3. Example:

1.3.1. Perhaps the best way to illustrate why AFSC uses AoP is to look at a simple example. Consider Joe who has an automotive repair shop. Joe has only one employee, himself. Joe currently has two customers who want him to repair their cars. He has a labor standards manual that shows how many hours it takes to make each repair. Joe uses those standards to give his customers their price. If Joe can get the work done in the length of time allotted, he makes money. If he takes longer, he makes less or could even lose money. If he can get it done sooner than the standard, he has created extra capacity so he can get additional customers or perhaps hire an additional mechanic.

1.3.2. Since Joe is the only mechanic, it is obvious that he cannot work on more than one thing at a time. But, to make his customers happy, he brings both cars into his shop and starts working on them. Joe spends 30 minutes finding all his tools that were not put back up after the last day's work. He works on one car for a few minutes and then moves to the next car. The first car requires a water pump, so he runs to the parts store to get one. When he gets back, he puts the water pump on the table and starts to work on the second car. This car needs a temperature switch. He again runs to the parts store to get the switch. When he gets back, he begins to put the water pump on the first car. After he gets it installed, he realizes he needs a new belt, so back to the parts store he goes. It does not take a scientist to realize that Joe has a problem. Joe ultimately takes two days to repair both cars. The standard says that he should have taken four hours each or eight hours total. Joe realizes that he cannot survive if he continues to work this way.

1.3.3. Joe decides to start using AoP to manage his work. He begins by setting a goal for himself. His stretch goal is to have every customer's car completed in 90% of the amount of time the labor standard says it should take. His next step is to map out the steps in the process so that he can better understand what is involved. Once he has the flow defined, he establishes some buckets of work in the process so that he can have milestones to help him monitor and communicate his progress. Included in those buckets are some triggers to let him know when he needs to have his

parts available. He realizes that since he can only work on one car at a time, he will only bring one car into the shop at a time.

1.3.4. Joe soon discovers there is a serious problem. He has a constraint. His constraint is himself. To meet his goal, he must be working on the car all the allotted time. How can he get parts if he needs to be working on the car the whole time? To exploit his constraint, he must find a way to get his parts delivered. He contacts the parts store and shares what his goal is and works out an agreement that they will deliver his parts to him within 30 minutes of order. In addition, Joe understands that to protect his available time for the next car there are some things that must be done before the next car can come into the shop. He sets up some criteria that establish all the things (like putting his tools in place) that must be done before the next car is ready to come into the shop.

1.3.5. Joe has just completed a very simple application of AoP. He has implemented some rules that will protect the flow of work through his shop. He identified his constraint and applied the rules to exploit his available capacity. His first pass may not result in his 90% goal, but he can then use tactical management tools to help continue moving toward his goal.

1.3.6. In Joe's example, the process is simple, and the solutions may readily be seen without applying the AoP methods. However, in more complex processes it may not be as obvious or the WIP may not be as easily seen as a car in a shop. This is the reason for the use of AoP. It allows organizations to simplify and create visibility in complex systems. AoP gives proven principles that allow the identification of constraints and ways to resolve them.

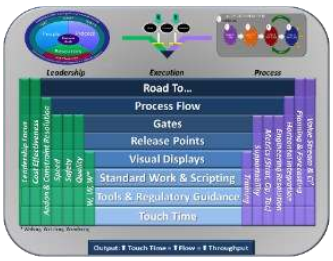
1.4. Key Components of AoP: AoP revolves around several key concepts and models. This handbook will explore each in more detail in the following chapters, but they are introduced here.

1.4.1. Leadership Model: The Leadership Model is a framework designed to guide leaders in achieving optimal performance and fulfilling the organization's mission. The AFSC Leadership Model provides a roadmap for leaders to optimize their operations, empower their workforce, and ultimately deliver exceptional results in support of the Air Force's mission.



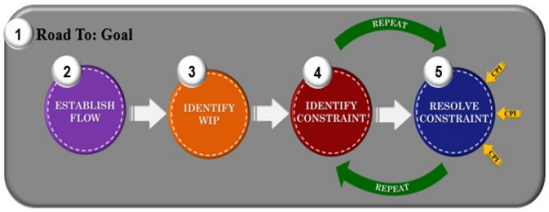
1.4.2. Science of Throughput: AFSC has embraced proven science in our management system and approach to achieving optimal throughput in their operations, drawing heavily on ToC. AFSC leverages the power of ToC and a dedicated focus on enabling flow to achieve operational excellence and meet the ever-evolving demands of Air Force sustainment.

1.4.3. Radiator Chart: The AFSC Radiator Chart is a comprehensive visual tool designed to guide the implementation of AoP methodology.



The Radiator Chart provides a holistic framework for AFSC to achieve operational excellence by aligning leadership, processes, and continuous improvement efforts.

1.4.4. AoP Cycle: AoP operates on a continuous cycle of improvement, focused on achieving organizational goals by optimizing processes. By repeating this cycle, organizations can continuously identify and resolve constraints, leading to a smoother, more efficient flow of work and ultimately achieving their desired goals.



1.5. Summary: While the complexities of our processes can pose challenges, AoP provides a framework for simplifying these complexities. By breaking down intricate systems into manageable components, AoP empowers analyzing, communicating, and continuously improving processes, ultimately leading to enhanced throughput, efficiency, and overall success.

Chapter 2

LEADERSHIP FOCUS

2.1. Introduction: This chapter explores the crucial role of engaged leadership in achieving AoP performance within AFSC. Positioning leadership as the cornerstone of a thriving AoP environment, the chapter examines the importance of common goals, effective management of people, processes, and resources, and the cultivation of a positive leadership culture characterized by teamwork, accountability, respect, transparency, credibility, and engagement. By understanding and applying the AFSC Leadership Model, leaders at all levels can empower their teams to achieve unprecedented success in sustaining weapon system readiness and generating airpower.

2.2. The Importance of Leadership: Leadership is more than a title or a position. It starts with creating a vision and developing a strategic roadmap to achieve the future state. It is also an art; the art of inspiring others to achieve common goals. It is not about checking boxes or wielding authority but about cultivating the right mindset and behaviors. Effective leadership requires continuous growth, practice, and a clear vision. It means influencing others through persuasion and collaboration, not coercion. Good leaders empower their teams to willingly contribute their best, using data and facts to build consensus and drive action.

A leader's potential is directly linked to the strength of their team. No one achieves greatness alone. Success depends on the collective efforts of those closest to the leader. Effective leadership begins with managing people, processes and resources to accomplish the mission. This includes communicating goals clearly and providing the necessary support for success, but true leadership transcends management. It requires developing individuals into future leaders and constantly refining processes to maximize efficiency. Only then can a team reach its full potential—the "Art of the Possible."

Servant leadership best exemplifies this ideal. It is a leadership style that prioritizes the needs of the team, empowering individuals to grow and perform at their highest level. Developing employees into leaders is crucial for achieving this "Art of the Possible." It demands intentional focus by engaged leaders who actively invest in both managing resources and cultivating new leaders. Engaged leadership is proactive, not reactive. It means advocating for necessary resources, fostering a sense of urgency, and rejecting complacency. Engaged leaders set high expectations for everyone, from internal teams to external partners, promoting collaboration and a shared understanding of the mission. They create a culture of problem-solving, where constraints are identified, addressed, and resolved efficiently. By continually motivating their teams to overcome challenges, engaged leaders ensure the mission consistently moves forward.

2.3. The Leadership Model: AFSC has developed a model to help communicate the essentials a successful leader must understand and execute to achieve true AoP performance. The model represents what the culture should embody. The right leadership culture will produce an environment for success. The culture of an organization is formed by the behaviors, values, and attitudes of the leaders; the culture then influences the



behaviors of each individual, and the behavior of the individual determines the level of success of the mission. Leaders are required to be fully engaged and create an environment which enables success. "Sustaining weapon system readiness to generate airpower for America" is AFSC's mission and overarching focus. Therefore, it is the duty of each leader at every level to embrace the Leadership Model and strive to understand and use it to accomplish the AFSC mission.

2.3.1. Leadership Model: The Leadership Model is built from the center out. The success of a leader begins with understanding and brokering the common goals of the organization and then engaging the levers of people, process and resources to successfully execute the mission. The mission is only executed successfully if the components of speed, safety, quality, and cost effectiveness are maintained in concert with common goals. The result of successful leadership is a culture where the mission is consistently executed and the traits of teamwork, accountability, respect, transparency, credibility and engagement are evident throughout the organization.

2.3.1.1. Common Goals: Setting goals is a fundamental part of success, but simply having goals is not enough. The true power of goal setting emerges when those goals become common goals—shared across all levels of an organization and aligned with partner organizations. Without this alignment, even the best intentions can lead to inefficiencies, miscommunication, and conflicting priorities. A unified vision ensures every team, from leadership to frontline employees, is working toward the same overarching objectives, fostering cohesion and driving meaningful progress.



Common goals should be the rallying point for every stakeholder united in executing a critical process driving behaviors that are focused on successfully executing the mission. To be truly effective, these goals must be defined in each of the critical components of success: speed (throughput), safety, quality, and cost effectiveness**.** If any of these components are not met, the mission is not fully successful. Furthermore, if a goal is not clearly defined and understood at every level, success cannot be measured. By aligning around common goals**,** organizations create a culture that delivers world-class results—quickly, safely, with high quality, and at the most effective cost.



For common goals to drive meaningful action, they must be broken down into measures and metrics relevant to each level of the organization. Senior leaders may focus on weapon system due date performance, while middle management may emphasize efficiency and throughput, and frontline employees may be measured by quality, safety, or earned hours. Similarly, in a complex enterprise like AFSC, every organization must understand its specific goals in relation to achieving the mission. It is imperative both leaders and the workforce understand their specific work center goals, as well as the role each individual plays in meeting them.

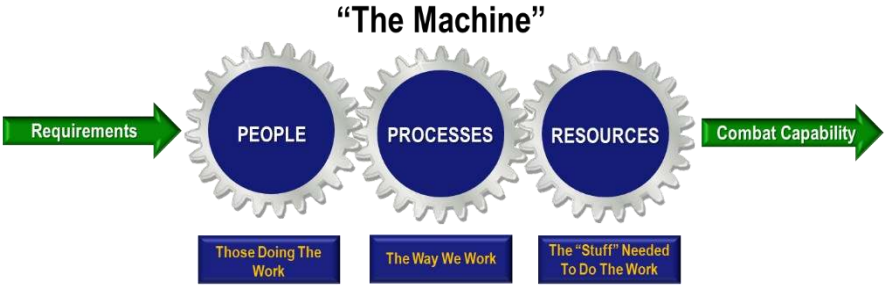
The same principle applies across partner organizations. If each entity is pursuing separate objectives without a shared set of measurable outcomes, collaboration becomes strained, and opportunities for synergy are lost. However, when common goals are collectively established and defined in a way which aligns measures and metrics at each level—such as a depot and a logistics provider agreeing on Key Performance Indicators (KPIs) like availability and timely delivery of parts and materials to meet real world readiness needs, each organization can optimize its operations toward a mutually beneficial outcome.

KEY CONCEPT: COMMON GOALS

Alignment Starts Here. For common goals to be effective, they must be decomposed from the enterprise level down to the individual shop floor. Every team member must understand how their specific work contributes to the larger target. Establishing a clear common goal provides a shared sight picture for the entire organization to measure execution and unite the team thereby establishing a path to determine the Road to… Goal.

It should not be expected that everyone in an organization be able to recite a list of organizational goals, but everyone should clearly understand what is expected in their work area and how they measure up against specific targets. This clarity allows teams to track progress, adjust strategies when needed, and ultimately ensure commitments are met. By fostering common goals at the right level of detail for every stakeholder, organizations create an environment where collaboration thrives, decision-making becomes more strategic, and success is amplified. The most effective organizations do not just set goals, they define them in ways that drive measurable progress at every level, ensuring every team and partner is moving toward the same destination, together. Common goals take on special relevance at the operational level. Operational leadership will be discussed in more detail in chapter 4.

2.3.1.2. People, Process, and Resources: Leaders are responsible for accomplishing a successful mission by operating a well-oiled machine of people, processes, and resources. Think of these three components as cogs in a machine. A leader is the operator who is constantly monitoring and adjusting the function of each cog to give the desired output, transforming a customer requirement into combat capability. The concepts of this machine will be developed in much greater detail throughout the remainder of this book.



Consider a builder who has a contract to build a house. He must first receive the requirement in a set of plans which define the details of the contract. Those plans will define the timeframe desired,

the available budget and the quality of the workmanship and materials to be used. The builder then must secure the funding, hire the right people and determine the processes to be used to safely achieve the desired quality. The builder then choreographs the people, processes and materials to produce a completed home for the customer within the agreed timeframe and budget.

The same is true of any organization in AFSC. People, processes and resources are the levers the leader has in his arsenal to achieve the mission. A successful leader must choreograph the use of those levers to achieve the defined mission. The mission delivery must be quick, safe, of high quality and cost effective.

2.3.1.2.1. People: The strength of AFSC lies in the dedicated, competent, and professional workforce. Without them nothing is accomplished. For this reason, leading people is an essential role of the manager in AFSC.



Every Workforce Needs:
Right Skills
Training
Education
Experience
Effective Leadership
Empowerment

A central role of a leader is to ensure those under their leadership are fully engaged in accomplishing the mission. Gallup defines engaged employees as: *those who are involved in, enthusiastic about and committed to their work and workplace*. Gallup. (2017). State of the American Workplace. Engaged employees do not happen by accident. The leader is responsible for communicating the goals to the workforce. Leaders must also define and communicate the role of every person in the accomplishment of the goals. The leader must provide ways for the individual process doer to understand their progress to achieving the target. The process doer needs the opportunity to develop skills to improve mastery of the job and to believe they have autonomy in achieving their own career goals. The leader is responsible for empowering the employee to be successful by ensuring the needed processes and resources are in place.

Availability of the correct number of people with the correct skill sets is elementary to success of the mission. The appropriate planning tools must be used to understand the requirement and communicate the need through the chain of command. While complete control of this process does not always exist at every level, the leader is still responsible for actively participating in the process. They cannot just rely on the system to give them their fate.

2.3.1.2.2. Process: To have engaged employees, they must be empowered through robust processes to perform the work. Effective leaders ensure all processes under their control or responsibility are current and complete. Processes should be standardized to the greatest extent possible and have predictable and measurable results. Leaders must engage direct reports in finding ways to improve their processes and be their champion to see those improvements acted upon. They must solicit the help of CPI teams or upper-level leaders where they do not have the expertise or authority to enact the improvements. Effective leaders will review the standard work of their direct reports often and

observe it in action to ensure it is being followed. They must ensure all processes are viewed with the end in mind. The goal of every process is to provide value by meeting common goals. If there are gaps or if the outcomes are not as expected, then the leader must recognize the constraint to reaching the goal and engage the AoP system to resolve the constraint.



Value Stream Analysis
Standard Work & Scripting
Rapid Improvement Event
Practical Problem-Solving
Model (PPSM)

Robust processes are not just a requirement for the direct report. Leaders must seek out and follow standard work for themselves. They must commit their actions to it. Where improvements are needed, they must engage the appropriate people to see that the improvements are implemented. Effective processes are not just checklists to be complied with but are actions that will lead to success. Great leaders seek ways to maximize their time, and the time of their employees, to create value and achieve the common goals of the organization. Leader standard work must be living and vibrant. All processes must be reviewed continuously to ensure they are up to date and followed correctly, so the processes are achieving the desired outcome.

2.3.1.2.3. Resources: Proper planning and responsible stewardship of resources is an essential prerequisite for success. Leaders are accountable for planning the right work environment and must identify needs lead-time away. Without proper planning and management of facilities, infrastructure, IT systems, equipment, tools, funding, and parts, mission capability and readiness can be seriously jeopardized. To achieve true AoP, leaders must do everything possible to maximize the value of the effort people contribute to the mission. Time spent searching for resources or finding work arounds reduces speed, creates safety concerns, impacts quality, and adds to the cost of the product. A commitment must be made to provide the workforce with the essential resources to safely achieve the mission and meet quality standards. However, budgetary constraints and real-world conditions may limit the ability to provide all desired resources.



Facilities
Infrastructure
IT System
Equipment
Tools
Funding

2.3.1.3. Mission Execution (Speed, Safety, Quality, and Cost Effectiveness): The inner circle of the model represents the components of successful mission execution. The end goal of every leader in AFSC is mission execution. The previous section discussed success is accomplished through the application of people, processes, and resources. But how do you know when the mission is a success? The mission is only successful if the common goals in the areas of speed, safety, quality and cost effectiveness are each met. Each one will be discussed briefly.



2.3.1.3.1. Speed: Speed is NOT about cutting corners or simply working harder and faster. Instead, speed is enhanced by the ability to quickly identify, elevate, and resolve constraints to the process flow. The workforce must feel constraint and issue resolution is a valued action. Maintaining critical path timelines demands the same level of urgency and focus as preventing wartime mission failure. Speed is about increasing throughput by application of proven scientific concepts to reduce flowtimes. Speed is enhanced when leadership embraces a safety culture, thereby reducing injuries and damage. Speed is also achieved when attention to quality eliminates rework. Enhanced throughput will ultimately lead to improved cost effectiveness. Greater throughput will lead to better customer satisfaction, greater capacity, and enhanced support of the AFSC mission.

Everyday interactions with processes in each person's personal lives demand speed. Everyone wants to be served quickly and accurately. Whether it is in a drive through at the fast-food restaurant, in the waiting room of the doctor's office or in the checkout line at the grocery store, everyone wants to be served quickly. Future choices are influenced by those experiences. The same is true with AFSC. If speed is pursued in everything, value will be created for every customer. Science confirms speed is directly linked to safety, quality, and cost effectiveness. The combination cannot be separated.

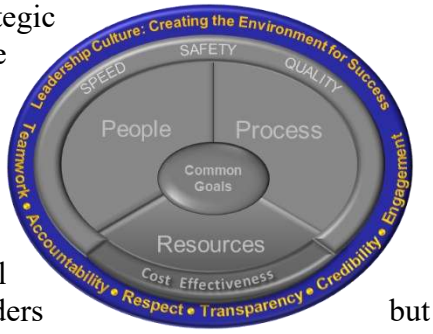
2.3.1.3.2. Safety: Safety is the priority of everyone, especially the leader. Zero mishaps and zero near misses are a possibility and are to be a goal of every organization. Leaders need to ensure everyone who comes to work for their shift goes home after their shift ready to give their best the next day. Safety is about application of the three machine components of people, processes, and resources by taking care of people and ensuring their work environment and processes always result in safety. Keeping every member of the team safe is critical to the success of the organization. Safety is also about protecting the speed, quality and cost effectiveness of the mission by protecting equipment, facilities, and products from damage. Damage to facilities, equipment, and products or injuries to personnel are all detrimental to the mission.

2.3.1.3.3. Quality: Quality is paramount. Defects in products have the potential for disastrous effects on the warfighter. Leaders reinforce the mandate for quality and take the necessary steps to ensure quality is primary. Trust and confidence are built by doing the job right the first time. Ford Motor Company championed the mantra "Quality is Job One." The idea this slogan represents is production without quality is not production at all, but rather mission failure. Mistakes will happen, but tools are available to identify and prevent repeats and take proactive steps to eliminate opportunities for error. Quality production leads to greater throughput, higher safety, and enhanced cost effectiveness. As AoP principles are applied, such as eliminating multi-tasking and reducing WIP, quality will improve.

2.3.1.3.4. Cost Effectiveness: The defense environment demands constant adaptation, requiring Air Force leadership to prioritize operational effectiveness in a resource-conscious manner. While mindful of spending, the primary objective is to ensure optimal mission success. To achieve this, a comprehensive understanding of the resources required to deliver end items is crucial. This understanding allows for strategic resource allocation and the elimination of inefficiencies. The taxpayer and the warfighter rely on the AFSC to provide available, affordable, and capable weapon systems on time and on cost. Enhancing the effectiveness of weapon system sustainment directly strengthens national defense capabilities. A relentless pursuit of speed, safety, and quality forms

the foundation for operational excellence. Shortcomings in any of these areas hinder overall effectiveness and compromise the readiness of the Air Force.

2.3.1.4. Leadership Culture: The outer ring of the Leadership Model represents the desired AFSC culture, along with the character traits that should define it. The focus on a culture of "leadership" is imperative because setting the stage for AoP results will come only through leadership focus. To put it simply: leadership matters. It is the tie that binds strategic planning with mission execution and makes it possible for the components and tenets of the Leadership Model to unite with the common goals needed for success. Creating the environment for success is the ultimate responsibility of leadership across the AFSC enterprise. The bottom line is if teamwork, accountability, respect, transparency, credibility, and engagement are present, an environment for success will exist. These traits do not characterize just the culture traits of leaders but the shared culture of all AFSC.



2.3.1.4.1. Teamwork: Cooperative and coordinated effort on the part of a group of people acting together as a team in the interest of a common goal.

2.3.1.4.2. Accountability: Expectations are communicated, and corrective action is taken when those expectations are not met.

2.3.1.4.3. Respect: Positive appreciation and consideration for the value of teammates demonstrated through specific actions and conduct indicative of holding them in high regard.

2.3.1.4.4. Transparency: Open and honest communication both horizontally and vertically.

2.3.1.4.5. Credibility: Believable and worthy of trust both individually and organizationally.

2.3.1.4.6. Engagement: Measure of an employee's emotional commitment to their leader, team, and mission.

2.3.2. Process Ownership: Taking Responsibility for the Entire Process

Every process must have an owner, but in practice this role is not always assigned, labeled, or formally acknowledged. In many cases, the correct owner is simply the leader whose mission depends on the outcomes of the process. Ownership, therefore, is not something bestowed by organizational charts; it is something recognized and accepted. The leader who realizes, *"The results of this process fall on me,"* is the one who must step into the role—regardless of whether they control every step within it.

Process ownership begins with understanding the entire Process Machine. This means seeing where work begins, how it moves through the process, where it slows down, and how the team impacts the overall performance. It requires paying attention to delays, handoffs, uncertainties, and issues that disrupt flow. Even when portions of the process fall under different functions, the owner must take responsibility for bringing the right people together, coordinating support, and removing

barriers as they arise. Their commitment ensures the process behaves as a unified system rather than disconnected parts.

Because ownership is rooted in leadership rather than authority, it depends on a mindset rather than a mandate. The right owner is the leader who understands that mission outcomes cannot be achieved without stewardship of the process that produces them. This leader accepts responsibility even when the process crosses boundaries, even when influence must replace authority, and even when solving problems requires persistent engagement with partners outside their span of control. Their leadership sets expectations for discipline, follow-through, communication, and accountability.

2.3.2.1. Characteristics of a Leader Who Naturally Becomes the Process Owner
A leader who steps into true process ownership is someone who:

- Recognizes that the success or failure of the process directly affects their mission.
- Is willing to accept responsibility even without explicit authority.
- Understands how the whole process works, not just their part of it.
- Actively works across boundaries to coordinate support and resolve issues.
- Approaches the process as a system rather than a sequence of siloed tasks.
- Models disciplined leadership
- Encourages continuous improvement

Process ownership is therefore not a title—it is a recognition of responsibility. It blends the practical work of monitoring the process with the philosophical conviction that leaders must take responsibility for outcomes they depend upon, even when they do not control every contributing element. When leaders adopt this mindset, processes become more stable, more reliable, and more capable of improvement because someone has finally taken responsibility for the whole machine.

2.4. Summary

Effective leadership establishes clarity of purpose, defines behavioral expectations, and cultivates a culture in which teams operate with confidence and discipline. When leaders communicate with precision, remain actively engaged, and accept responsibility for outcomes, they align people and actions in support of mission achievement.

Within this leadership framework, genuine process ownership becomes essential. The leader who relies on the output of a process must assume responsibility for the performance of the entire process, even when specific elements fall outside their direct authority. By understanding how work progresses, addressing obstacles promptly, and coordinating efforts across organizational boundaries, leaders ensure the process functions as an integrated system. This level of ownership strengthens organizational culture, enhances stability, and establishes the foundation for sustained improvement.

CHAPTER 3

SCIENCE OF THROUGHPUT

3.1. Flow

AoP represents the application of established scientific principles to radically enhance the AFSC mission, pushing the boundaries of what was once deemed possible. The typical measure of mission success is the output of the organization described in how many units or "things" the organization produced in a particular period (i.e., aircraft per year, contracts per year, or personnel actions per month). In the lexicon of AoP, output is synonymous with throughput. Often the phrase "Throughput is King" is used to emphasize the importance of achieving high levels of throughput. While throughput is the count of how many "things" emerge at the end of the process, movement of material or "product" through the process is referred to as flow. Maximum throughput is the result of achieving rapid uninterrupted flow in the process. Therefore, achieving AoP results hinges on achieving maximum flow rate of work through the process.

KEY CONCEPT: THROUGHPUT IS KING

Throughput is the required output of a production machine expressed in units per time. The entire science of the production machine is built to enable and increase throughput. Remember: For a constant throughput, increasing your speed (reducing flowtime) will reduce Work-in-Progress (WIP), freeing up critical resources and capacity.

When flow is streamlined and constraints to flow are resolved, the result is greater throughput. If the input rate remains constant, then increased throughput will reduce WIP. With reduced WIP comes reduced resource requirements - less dock space, less shop space, less equipment, less labor costs, and less supporting overhead. Focusing on throughput and the supporting measures and metrics provides the mechanism that will lead to reduced costs and increased capabilities for the Air Force. The most important result of flow is that it allows us to reach stretch goals and positions the customer and the organization for the future. Every AFSC organization should strive to create movement through the process that is a steady and continuous stream. Flow should be free from disruptions, where value is being added to the product as it moves quickly through to output. Continuous uninterrupted flow will naturally lead to higher quality, greater safety, and less cost.

When is Throughput not King?

There are times when the primary objective of a process may not lend itself to the typical metric of units over time. An example may be when a process's purpose is to deliver an annual budget or forecast. In this case, the metric of success may present itself in quality of the product instead of the typical measure of units over time.

3.1.1. Disruption of Flow: All work is a process, and every process has flow. Unfortunately, there are usually inherent enemies that impact flow and thereby negatively affect the throughput of the organization. Each process is a composite of numerous sub-processes, each with its own set of issues that potentially impact throughput. It is easy to imagine how issues can drastically reduce the success of the organization. There is a common adage that states improvement cannot occur until a problem is acknowledged. The same holds true for improving the throughput of a process. These potential issues must be embraced and recognized in order to take appropriate action to protect the flow of the process.

The enemy takes many forms and can attack from all directions. The following paragraphs will discuss some of the most prevalent issues that work together to reduce success. Some are obvious, but others are not. However, when all are acknowledged, it becomes possible to develop a plan for a counterattack.

3.1.1.1. Process Constraints: The first enemy to flow is process constraints. Eliminating process constraints is the primary focus of AoP. Imagine a garden hose with water flowing through it. The hose, the faucet, and the spray attachment on the end of the hose constitute the process. A single component of that process is the determining factor that governs how much water can be delivered to the plant. This limiting factor is called the constraint. It may be the faucet, the size of the hose, a kink in the hose, or the nozzle size of the sprayer.

To maximize the amount of water provided to the plant, the gardener must analyze the process to determine what must be adjusted to improve water flow. If the hose is kinked, opening the nozzle wider will not improve the flow. The kink must be removed before flow will improve. While this example may seem overly simple, every process has a constraint or weakest link that must be resolved before the throughput will improve.

The tools of AoP provide the means to identify those constraints that disrupt flow. A systematic methodology to resolve constraints will be discussed later, but the first step to improvement is understanding that a constraint exists in every process and must be found before progress can be made.

3.1.1.2. Parkinson’s Law: Parkinson’s Law states, “the time it takes to perform a task will expand to fill the time allotted.” If two days are allowed, two days will be used, even if the task could be completed in a half-day. However, few are willing to acknowledge wasted time or the stretching out of work.

Even when time is not consciously wasted or procrastination is avoided, there is a tendency to add finishing touches that may not be necessary. This behavior is a reality of human nature. All individuals are susceptible, and it is a significant enemy to rapid and continuous flow through the process.

3.1.1.3. Student Syndrome: Student Syndrome is a close cousin to Parkinson’s Law. This principle reflects the common behavior of delaying the start of tasks until close to the deadline. This is another reality of human behavior that is often unacknowledged, but it is very real.

The name originates from common academic behavior. For example, an assignment is given on Wednesday to be completed by the following Monday. Based on experience, the task can usually be completed in a couple of hours. Many activities fill the week, and the remaining time appears sufficient. Finally, on Sunday night, the task starts because no more time remains. Student Syndrome is a deadly enemy to flow and throughput.

3.1.1.4. Natural Causes: No system is perfect, and every process is subject to natural variation. For example, more discrepancies may be discovered during inspection than expected. Aging systems may experience more failures than predicted. Troubleshooting or debugging may take longer than anticipated.

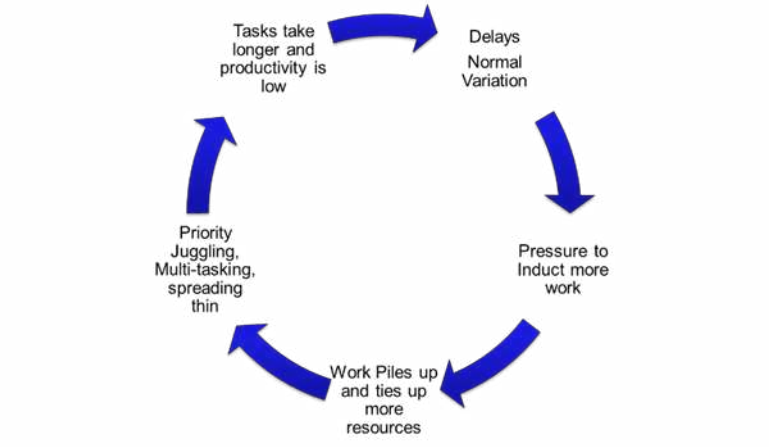
Additional factors such as funding shortfalls, insufficient information, or lack of skill proficiency can also delay progress. In other words, things often do not go as planned. These unplanned disruptions can cascade through a system, compounding delays and obstructing flow. Such disruptions are intrinsic to all systems and are not easily removed.

Intrinsic disruptions to flow will inevitably cause delays. When these delays are combined with the effects of Parkinson's Law and Student Syndrome, throughput can suffer exponentially.

3.1.1.5. Self-Inflicted Causes: In real-world systems, many disruptions to flow stem from the way natural delays are handled and managed. When delays occur and commitments begin to slip, there is pressure to start or release work earlier than advisable to meet deadlines.

Inducting work too early, or before it is supportable, causes work to pile up in front of resources, resulting in increased wait times and blocked flow. As wait times grow and delays increase, urgency mounts, leading to priority conflicts and multitasking.

Resources are then spread thin across more jobs. Each job receives less attention than necessary and therefore takes longer to complete. Supporting personnel and partners must also manage more issues and respond to shifting priorities.



The net result of multitasking, priority conflicts, and overextended resources is that tasks take longer than planned and productivity suffers, creating even more delays. This vicious cycle of delays generating further delays is a key source of low throughput.

The pressure to induct work prematurely often stems from the assumption that all individuals must remain constantly busy and that an earlier start leads to a quicker finish. Unfortunately, in systems with limited capacity, starting work earlier increases the volume of WIP.

High WIP leads to longer wait times, greater priority conflicts, increased firefighting, more multitasking, and further dispersion of resources, all of which produce more delays and reduce throughput. And thus, the vicious cycle begins. Excess WIP is the enemy of flow.

3.2. Theory of Constraints (ToC):
When all the inherent enemies to flow are considered, it may seem like a no-win situation. Constraints, human nature, and natural variations in work cause delays and typical reactions cause even more delays. The good news is, there are principles and actions that drastically improve flow and make managing the process more achievable. This handbook provides methods to apply the concepts of ToC to various types of work.

3.2.1. ToC Focusing Steps: In the book "The Goal" (Goldratt et.al., 1997) the author lays out the foundation behind the ToC approach to process improvement and creation of flow. ToC establishes a series of focusing steps that are used to eliminate constraints to flow. These steps will apply to every process, both production and administrative. The premise of ToC is that processes act much like a chain that is only as strong as their weakest link. The weakest link in any operation is described as the constraint, which prevents the output of the entire system from meeting the desired performance. The constraint is usually identified as the process or resource that has the most WIP, or the most heavily loaded process or shop in the system. The utilization of ToC requires all CPI resources to be focused on the constraint. Just as opening the nozzle wider will not improve flow if the garden hose is kinked, CPI efforts that do not attack the constraint will not improve the overall system performance and could make the entire system perform less effectively. The five basic and sequential steps for proper application of ToC are described below.

3.2.1.1. Identify the Constraint: As mentioned earlier, the constraint is usually identified as the segment with the lowest throughput rate. This may manifest itself with queue before the segment or excessive WIP in the segment due to poor performance.

3.2.1.2. Exploit the Constraint: Usually this involves obtaining the immediate maximum potential out of the constraint without significant investment. For example, if a process tool limits the system's output, the constraint is exploited by continually operating the process during second or third shifts, lunch breaks, weekends, etc. Buying a second process tool is not a way to exploit the constraint. If people are the constraint, then having all parts, tools and materials available to maximize their touch time on the process will create the most value for the customer.

3.2.1.3. Subordinate everything to the constraint: It is inefficient to allow unconstrained functions to operate above the limiting constraint's capability. All operations must match the constraint's pace to prevent overwhelming or starving the preceding and succeeding tasks. Support functions must also be subordinate to the constraint. Competing priorities must be resolved so supporting resources can devote time and energy to protecting and resolving the constraint. This is an area where horizontal integration is necessary so that all stakeholders have a clear understanding of the common goals.

3.2.1.4. Expand or Elevate the Constraint: Most constraints will be resolved by the time step three has been completed, but in some cases, it may require elevation of the problem. Expansion of the constraint is elevating the output of the limiting process resource or shop until overall system performance can be met, or until another constraint becomes the limit to the system. This might include the purchase of additional equipment, addition/realignment of personnel, or preferably, an increase in output through waste removal or other CPI activity.

3..2.1.5. Repeat the Process (Identify, Exploit, Subordinate, Expand): If process performance is still less than the needed throughput, these steps must be repeated until all constraints are resolved. It should be noted that if the desired performance of the system is attained, no limiting constraint requires attention; however, different processes or shops may have different capacities (i.e., the system may not be balanced) or it may be time to re-evaluate the stretch goal for the future.

NOTE:

These five Focusing Steps are the foundation of a constraints-based management system. The application may be more direct in some processes than in others, but they all use the same premise to achieve flow:

1) Identify the constraint.

2) Exploit the constraint to get the most out of it.

3) Subordinate all other processes and resources to the constraint.

4) Find ways to expand the capacity of the constraint. (If it has not been solved)

5) When the constraint is resolved move on to the next constraint.

3.3. Rules of Flow (RoF)

When Eli Goldratt first introduced ToC in 1997, the term "Rules of Flow" (RoF) did not exist. Over the years, organizations that have successfully implemented the various applications of ToC have discovered there are certain actions that ultimately lead to the most improvement. Over time, the ToC community has adopted the phrase RoF to encompass a set of universal principles. If true AoP type improvements are to be seen, these rules must be embraced and pursued at all costs. The rules vary from organization to organization based on their needs and circumstances, however there is commonality among them all.

Since AFSC began the AoP journey, a unique set of RoF have been developed that fit the needs and vocabulary of AFSC. Experience has taught that RoF is the common thread that runs through every highly successful organization. No matter how an organization applies AoP or ToC, passionate pursuit of RoF will lead to greater success. Whether an organization uses a traditional process (to be discussed later) or an administrative office needs to manage their daily workload, RoF is the answer to success. Application of AoP takes on many forms across AFSC, but each successful application shares the basic principles of RoF.

It may have been said (perhaps by you) "AoP does not apply to our process." Nothing could be further from the truth! Yes, it is true that a production-oriented process may not line up well with an administrative office, but RoF applies in every situation and in every process. Any organization that embraces RoF and utilizes the various tools presented in this handbook will be successful in achieving improvement. The following paragraphs provide an overview of RoF, but more insight will be provided in discussion of the application of the management of processes in chapter 4.

3.3.1. Control WIP: Excess WIP is the enemy of flow. WIP leads to priority conflicts, resource depletion, multi-tasking, long wait times, firefighting, and overall poor throughput. Healthy processes must define the correct amount of WIP that should be in the system and define where it should be located. Too much WIP in the process is one of those self-inflicted killers of flow. WIP must be minimized to the lowest level necessary to achieve the throughput goal. The next section will discuss "Little's Law" that helps an organization to define the right amount of WIP that should be in the process.

3.3.2. Release Only Supportable Work: The release control element of RoF communicates the discipline to only start work when it is fully supportable. Release control, as discussed here, is the "leadership resolve" to follow the criteria and demand discipline to the concept. Release control is focused on ensuring the process is fully supportable, and everything needed for work is available before being released to the doer. For a production process, this means all skills, parts, tools, materials and tech data are on hand, or there is data driven by high confidence it will be available prior to the need date. For an administrative process, this may mean all data is complete and all funding is available. AFSC processes having everything fully supportable with adequate personnel (with right proficiency levels & skills), parts, tools, equipment, work documents, troubleshooting tech data etc. before starting work on the job will significantly reduce flowtime. Managing WIP is a key enabler of full supportability and vice versa. Once WIP is controlled, it is important to release work only when fully supportable. Complete supportability provides everything needed at the point of execution/use, generating one overall start action and one stop action; thus, eliminating multiple starts and stops to gather parts and tools. On the flip side, when the work can be started and completed immediately, there is no pressure to induct additional work.

Example 1: When completing tax returns it is more beneficial to gather all the information needed before beginning. The taxes cannot be completed until all the information is available. It may seem beneficial to begin the filing process prior to receiving all the pertinent documents such as W-2s, mortgage interest reports or donation receipts. But, to do so requires certain steps to be skipped and increases the likelihood of omissions or errors. In the end, the return cannot be submitted until all documentation is available and all steps completed. Having all the needed documentation in hand reduces the time required to complete the process, improves quality, and reduces frustration.

Example 2: Imagine starting a do-it-yourself home improvement project. Is it better to have all the tools, parts and material on hand before starting, or to make multiple trips to the home improvement store after work begins? Extra trips have a significant impact on the flowtime of the project. Proper planning and preparation and having everything ready before starting the job enables the job to be finished quicker with better quality and much less frustration.

3.3.3. Assign Work Using a Synchronized Plan: When tasks are fully supportable and have passed all the business rules for release control, they should be loaded with the appropriate personnel to accomplish the job as quickly as possible. The work should be assigned based on the daily scripted work plan and priorities. Following a disciplined flow of work will allow the synchronization of enterprise resources to execute the plan. All players in the process should be in tune with the daily plan and maximize utilization of resources to meet the expectations. Proper assignment of work minimizes multi-tasking and allows the doer to Focus and Finish the task before moving on to the next task. Not only does this enhance the speed of the process, but it greatly enhances the quality of the product. The opportunity for quality escapes is minimized by eliminating starting and stopping and creating fewer handoffs. This synchronized plan should be a daily plan. Supervisors cannot hand out a week's worth of tasks on Monday and expect to be on track on Friday. Natural variation and disruptions to flow will happen and must be dealt with and the plan must be adjusted continuously.

3.3.4. Quick Issue Resolution: Quick issue resolution is foundational to the AoP management system. Without quick resolution of issues, the rest of the tools within AoP are meaningless. When issues arise that impact WIP levels or throughput, they must be dealt with quickly. When constraints affect the supportability of the task, they must be resolved. If personnel issues prevent proper resource loading, then find ways to resolve the constraint. AoP is about using the tools to discover constraints and then resolve them quickly. Quick issue resolution is the desired outcome of the Leadership Model. It is the job of every leader at every level. Leaders and supervisors should make frequent contact with process doers throughout the day to identify issues and ensure prompt resolution. While leaders may use the resources available to them to resolve issues and constraints, they cannot just delegate the responsibility to others. **If quick issue resolution is not carried out with a sense of urgency, the other components of RoF are just irritations that stop the flow while hoping for things to get better.**

3.4. Little's Law: The previous section discussed RoF and the number one rule, control WIP. To control WIP, there must first be an understanding of what the correct amount of WIP should be. For simple processes or many administrative processes, that may be an intuitive solution or one that can be discovered by trial and error. However, for more complex processes one must grasp the concept of Little's Law before the methodology behind WIP calculation can be understood. A description of Little's Law will help strengthen the understanding of important concepts such as throughput, flowtime, WIP, and takt time. Little's Law provides the foundation for creating and setting up a process.

Why is Little's Law important? After years of implementing AoP in AFSC, many organizations have established processes that have defined WIP levels. Therefore, the question may be if it is important to understand and use Little's Law. The answer is yes. Every supervisor needs to understand how to apply it so that they can constantly evaluate their processes. When demand changes or new workload is inducted, new processes need to be defined. Little's Law forms the basis for distributing WIP and setting target days throughout the process with an even flow from induction to output. At steady state, all processes have an average throughput, WIP, and flowtime. The fundamental relationship between all three is described by Little's Law: WIP = throughput x

flowtime. Throughput is the required rate of output of a process expressed in units per time (for example units per month). Flowtime is the average time that a unit stays in a process. WIP is the average number of units in work throughout the process. To fully understand the relationship between these three components (WIP, throughput and flowtime) and how they relate to AFSC's concept of speed, use math. The equation can also be expressed in terms of the desired target as shown below:

$$WIP = TP\ x\ FT \qquad TP = \frac{WIP}{FT} \qquad FT = \frac{WIP}{TP}$$

WIP = throughput x flow time

Throughput is the required rate of output of a process expressed in units per time (for example units per month). Flow time is the average time that a unit stays in a process. WIP is the average number of units in work throughout the process.

In AoP, increased speed equals reduced flowtime. For a constant throughput, reducing the flowtime will reduce WIP. If the system has unlimited demand, and a constant WIP is maintained, then reducing the flowtime will result in an increased throughput for the process. It is important to understand these relationships because focus on improving speed will result either in 1) reduced WIP or 2) increased throughput for the process, or both.

For the purposes of an AFSC process, Little's Law may be modified to include the concept of takt time. Takt time is the heartbeat of a process. It defines how often a single unit must be produced from a process to meet the stretch goal. For example, a takt time of 10 days means the process must produce one unit every 10 days. Mathematically, it is the reciprocal of throughput as defined above. Takt time is determined by dividing the available time (AT) by the required output (RO) in that amount of time expressed in units of time).

$$Takt\ Time = \frac{AT}{RO}$$

It is important to note that when calculating takt time, the available time for a process should reflect the total number of units of time that is available, whether it is in minutes, hours, days, months or years. The required output is a measurement of customer demand, or how many products or units of service a process-doer is required to complete in the given period of time that is available. For example, if a process is designed to produce 37 units in one year, the throughput rate is 37 units / 365 days or 0.1 units per day

$$37 units/365 days = \ 0.1\ units/day$$

The takt time would be 365 days divided by 37 units which are equal to a takt time of 10 days.

$$365 days/37 units = 10\ days/unit$$

Every 10 days the process must produce a unit, and all enterprise teammates must support this tempo. Another example would be to imagine a doctor's office that operates 600 minutes per day (10-hour shift) with a demand of 30 patients per day. The takt time is then calculated:

$$600 min/30 patients = 20\ min/patient$$

In other words, the doctor cannot average more than 20 minutes with each patient to meet the requirement of treating 30 patients per day. AFSC modified version of Little's Law then becomes:

$$Flowtime = WIP\ x\ takt\ time$$

When designing a process, two of the three variables in Little's Law must be defined. As an example, assume there is a stretch goal for a process to produce 64 aircraft per year with a limited WIP of only 23 aircraft. The production output requirement of 64 aircraft per year is defined by the future state customer requirement. The WIP target is defined by a) the customer [in this case the Aircraft Availability (AA) requirement or b) an internal goal to reduce cost and create capacity. The takt time is calculated by dividing the days available by the required output in that available time.

$$365\ days \div 64\ units\ = 5.7 \frac{days}{unit} = takt\ time$$

Every 5.7 days this process must produce an aircraft. Flowtime then equals WIP x takt time.

$$23\ units\ x\ 5.7 \frac{days}{unit} = 131\ days$$

This process must perform at a speed (flowtime) of 131 days to output 64 aircraft per year, while maintaining a total WIP of only 23 aircraft.

Available Time (Days)	Required Output	Takt (Days)		
365	64	5.7	23	WIP
	Req'd Flowtime (Days)		131	Cal. Days

Through the above example, it can be seen how the parameters of the process are designed based on a future state stretch goal. The resulting flowtime target is now understood, and the organization must do everything possible to achieve improvements in the process to achieve that goal.

Another important component of the formula is the overall capacity of the organization. For instance, the footprint of an individual weapon system, based on its AA calculation alone, may be greater than the physical space available to a complex given its total workload requirement. There may be cases where increasing the speed of a weapon system is necessary to reduce its footprint (WIP) to free capacity for new or increased workload for the complex. For this reason, it is important to understand the workload requirements of the organization in its entirety to ensure the individual stretch goal allows the organization to meet its overall workload obligations.

Caution:

Application of Little's Law does not have a cause-and-effect relationship to process performance. It is only a mathematical equation that shows the relationship of the variables in a steady state process. It provides the basis to calculate the foundational levels for the process and expresses the mathematical representation of the stretch goal. Progress toward the stretch goal is measured against this calculated target.

3.5. Typical Implementation Methods: It does not take a lot of experience to realize there are vast differences in the types of work that are performed in AFSC. For that reason, it should not be surprising that application of AoP can be significantly different depending on the type of work accomplished. This is not unique to AFSC. Organizations across the globe have tailored ways to apply ToC to achieve optimized throughput to match their situations. Each method is designed to address the inherent enemies of flow and maximize the throughput of the constraint. This section will give descriptions of various ways that AFSC has used ToC concepts in implementation of the AoP management system. While these applications are tailored to the organization, they all focus on maximizing throughput by exploiting the constraint through application of RoF.

3.5.1. Traditional Flow. A traditional flow process is a multi-step process that consists of a process flow with a critical path divided into buckets of work. Each bucket is then defined by calculated WIP levels and entry and exit criteria to control the transfer of work in and out of the bucket. WIP levels can be calculated using Little's Law for both the process and each bucket.

Consider Joe's garage discussed in the introduction. Joe's repair shop follows a simple process, demonstrating the key elements of a traditional process:

Segment 1: Inspection & Diagnosis – Identifies the required repairs before starting work.

Segment 2: Parts Acquisition & Preparation – Ensures all parts and tools are available before repairs begin.

Segment 3: Repair Execution – The core work process, where Joe completes repairs efficiently.

Segment 4: Final Inspection & Delivery – Verifies work quality before handing the car back to the customer.

Joe initially struggled with inefficiencies due to multitasking and not exploiting the constraint. By implementing traditional flow, he:

1. Controlled WIP by limiting work to one car at a time, preventing task switching.
2. Established release points to ensure all necessary parts and tools were available before starting each repair.
3. Protected his constraint (his own time) by having parts delivered instead of making trips himself.

These changes allowed Joe to streamline his workflow, reduce waste, and improve turnaround times. Joe's process aligns with RoF, which applies to any traditional process:

1. Control WIP – Work is limited within each step to maintain efficiency and avoid overload.
2. Release Only Supportable Work – Work enters the system only when all prerequisites (tools, parts, resources) are met.
3. Assign Work Using a Synchronized Plan – Work is scheduled and sequenced based on a structured plan.
4. Quick Issue Resolution – Problems are addressed proactively, such as securing parts delivery to prevent downtime.

3.5.2. Drum Buffer Rope (DBR): is a planning and scheduling methodology from the ToC. It is specifically designed to maximize throughput while managing variability in production processes. This section provides guidance on implementing DBR within an organization, ensuring alignment with RoF.

3.5.2.1. Core Components of DBR

3.5.2.1.1. Drum: The drum represents the system's constraint, which sets the pace for the entire process. Examples of constraints include critical processes, specific skillsets, or bottleneck tasks. To optimize system performance, the constraint must be effectively managed, as the overall throughput cannot exceed its capacity.

3.5.2.1.2. Buffer: Buffers protect the constraint from interruptions caused by variability. There are two types:

- Time Buffers: Ensure upstream processes complete their tasks on time.
- Inventory Buffers: Maintain a queue of resources (e.g., parts or materials) to keep the constraint continuously operational.

The goal of the buffer is to prevent idle time at the constraint, ensuring no lost capacity.

3.5.2.1.3. Rope: The rope acts as a signal to control the release of work into the system. It ensures that work is released at the same rate as the constraint's capacity, preventing excessive WIP and maintaining smooth flow.

3.5.2.2. Implementation Steps

3.5.2.2.1. Step 1: Identify the Constraint: Use process metrics to pinpoint the resource with demand exceeding its capacity. Or the resource with the lowest capacity if the customer demand is lower than the process capacity.

3.5.2.2.2. Step 2: Establish Buffers

- Place buffers strategically before the constraint.
- Use time buffers to protect schedules and inventory buffers to ensure the constraint always has sufficient resources.
- Continuously monitor buffer health and address fluctuations promptly.
- To safeguard the project's completion date, a project buffer is strategically added to the schedule to account for unforeseen issues and maintain the promised completion date.
- For accurate project tracking and auditing, both planned and buffer times for the total project critical path must be reported (accounting for in the machine).

3.5.2.2.3. Step 3: Create and Maintain the Rope

- Develop a clear signal system to release work into the process at a controlled rate.
- Ensure the rope aligns with the constraint’s capacity to prevent overloading or underutilization.

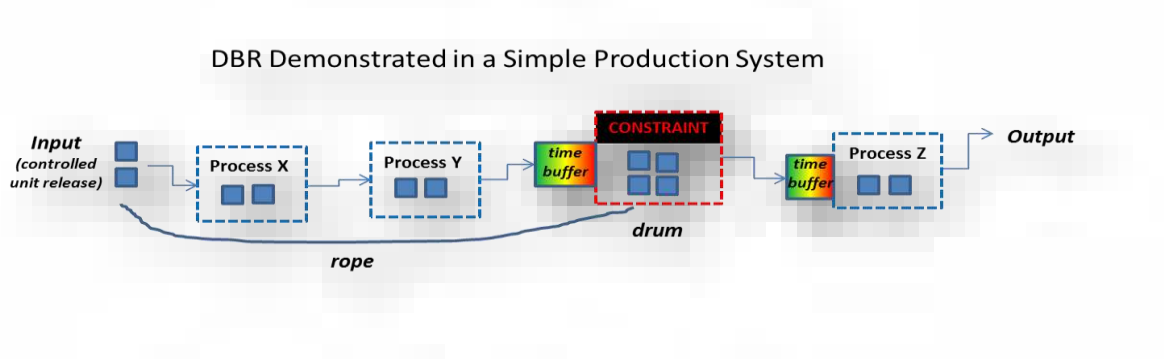
3.5.2.2.4. Step 4: Synchronize and Subordinate

- Align all non-constraint processes to the pace of the drum.
- Focus on maximizing constraint throughput while avoiding excess WIP or multitasking.

> In cases where the process has sufficient capacity to meet the current demand, the customer demand or market then becomes the constraint. In these situations, the segment with the lowest capacity should be chosen as the drum and managed to produce at the takt time of the process. This will facilitate WIP control and ensure the process operates efficiently without over producing.

3.5.2.2.5. Step 5: Monitor and Adapt

- Regularly review system performance and buffer conditions.
- Utilize tools such as Pareto analysis and Lean/Six Sigma to address root causes of variability.
- Adapt processes to ensure continued alignment with customer demands.



3.5.2.3. Best Practices

3.5.2.3.1. Constraint Management:

- Dedicate resources to optimizing the constraint.
- Prioritize CPI activities that enhance constraint performance.

3.5.2.3.2. Buffer Management:

- Continuously monitor buffers and adjust to accommodate process variability.
- Avoid excessive WIP, which can increase lead times and complexity.

3.5.2.3.3. Controlled Work Release

- Only induct work that is fully prepared (full-kit readiness).
- Use synchronized daily plans to guide work assignments.

3.5.2.3.4. Quick Issue Resolution:

- Identify and address disruptions promptly to maintain smooth flow.
- Empower teams to escalate and resolve issues quickly and efficiently.

3.5.2.4. Benefits of DBR

- Increased Throughput: Maximizes the productivity of the constraint.
- Reduced Variability: Proactively protects schedules from variability.
- Enhanced Responsiveness: Reduces lead times and improves system adaptability.
- Controlled WIP: Prevents system overload and optimizes resource utilization.

3.5.2.5. Key Take Aways:

By implementing DBR, organizations can achieve greater operational efficiency, align processes with customer demand, and enhance overall system performance. This structured approach will ensure continuous improvement and support commitment to excellence through controlled workflow and synchronized planning.

3.5.3. Focus and Finish: Focus and Finish is a subset of Simplified Drum-Buffer-Rope (S-DBR) used to manage WIP in processes typically completed by one person. It prevents process doers from taking on excessive tasks and fosters timely project completion by controlling the release of work. This method is especially effective in administrative areas where tasks are assigned based on responsibility, skill, or other factors.

3.5.3.1. Key Principles of Focus and Finish

3.5.3.1.1. Control WIP: All WIP is held in a queue, with only a limited number of tasks assigned at a time. The number of tasks assigned depends on the individual's bandwidth and the complexity of tasks.

3.5.3.1.2. Fully Supportable**:** Before any task is assigned, it must be fully supportable, meaning all necessary materials, information, and resources must be available. A subject matter expert should review each task to ensure that all are fully supportable. This ensures that tasks can be completed without delays or interruptions.

3.5.3.1.3. The Constraint is the Individual's Bandwidth: Each person's capacity to handle tasks effectively acts as the constraint in this system. Their available time must be exploited to maximize their time without wasting it doing non-value-added work. Work should align with their ability to complete tasks without excessive multitasking.

3.5.3.1.4. Finish Before Starting New Tasks: Assigned tasks must be completed before any additional work is assigned. This approach ensures efficiency and reduces the risk of delays caused by multitasking.

3.5.3.1.5. Applicable Across Various Processes: Focus and Finish is suitable for processes involving:

- Individuals or small teams
- A relatively small number of tasks
- Short-duration touch times

3.5.3.2. Steps to Implement Focus and Finish

3.5.3.2.1. Step 1: Establish a Work Queue

- Create a centralized system to hold all WIP.
- Assign a subject matter expert to determine if the work is supportable.
- Ensure tasks are fully supportable before they can be assigned.

3.5.3.2.2. Step 2: Define Bandwidth and Task Limit

- Assess each person's capacity to accomplish work efficiently.
- Set a maximum number of tasks that can be assigned simultaneously based on the nature of the work.

3.5.3.2.3. Step 3: Assign Tasks

- Assign tasks from the queue based on priority, skill alignment, and responsibility.
- Ensure everyone has no more than their defined task limit.
- Verify that each task is fully kitted before assignment.
- Tasks in work should be prioritized to a synchronized plan based on real world needs.

3.5.3.2.4. Step 4: Focus and Complete Tasks

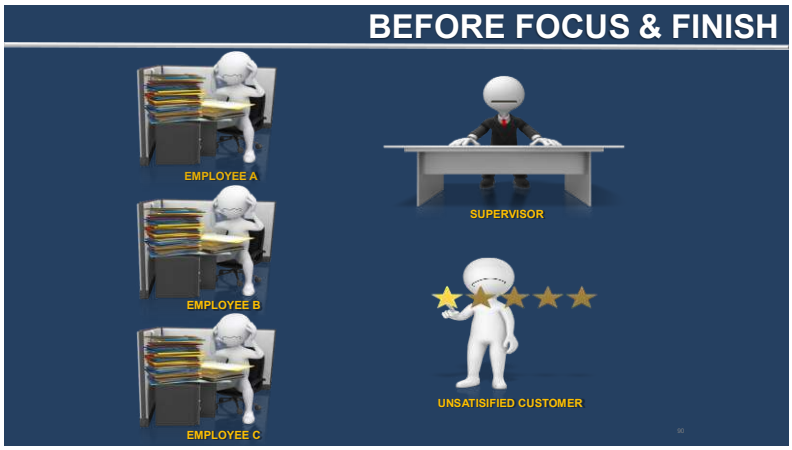
- Instruct process doers to focus exclusively on their assigned tasks.

- Avoid interruptions or reassignment of priorities mid-task unless critical.

3.5.3.2.5. Step 5: Release New Tasks After Completion

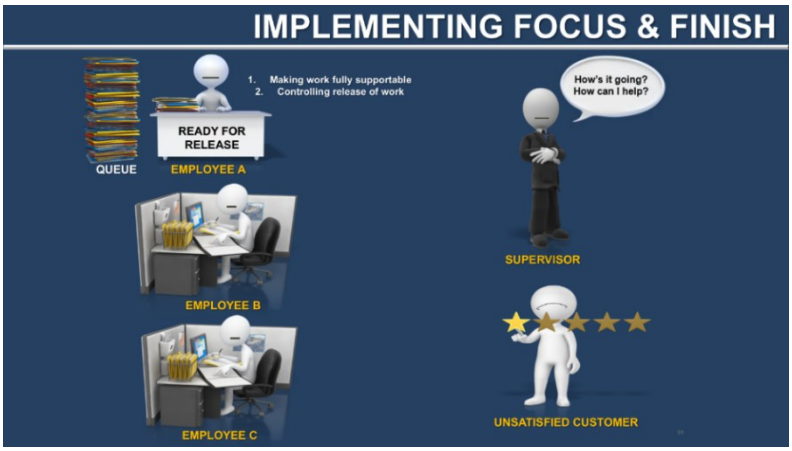
- Once all assigned tasks are completed, new tasks can be drawn from the queue.
- Repeat the process while maintaining WIP control.

Graphical Illustration of Focus and Finish implementation in an office environment.



Before Implementation

Excess WIP at every level, employees are stressed, and customer is unhappy.



During Implementation

Excess WIP is placed in queue, SME is assigned to make work supportable. Employees are less stressed, but customer is still unhappy.



After Implementation

Excess WIP is eliminated. Employees are not stressed, and customer is happy.

3.5.3.3. Tips for Effective Focus and Finish Implementation

- Clear Communication: Ensure all team members understand the system and their roles.
- Resolve Issues Quickly: Resolve any impediments to task completion promptly.
- Regular Monitoring: Review WIP levels, task assignments, and completion rates to identify bottlenecks.
- Adapt Task Limits: Periodically reassess task limits based on changes in workload or process efficiency.

3.5.3.4. Key Take Aways: Focus and Finish is a powerful method for managing WIP in individual-driven processes. By adhering to its principles, teams can reduce multitasking, increase efficiency, and ensure timely project completion. Regular evaluation and adaptation of the system will further enhance its effectiveness.

3.6. Summary

The practices outlined in this chapter establish the essential disciplines that enable processes to function as stable, predictable systems. Effective implementation requires clear standards, structured coordination, and consistent engagement with the work as it is performed. When leaders and teams understand the sequence of activities, monitor performance with rigor, and address issues at their source, the process begins to operate with reliable rhythm and reduced variation. These fundamentals form the operational backbone necessary for applying the principles of ToC, ensuring that work flows in a manner that protects the system's limited capacities and strengthens mission execution.

As these methods take hold, leaders must continue to view the process as an integrated machine rather than a set of isolated tasks. Sustaining alignment, maintaining steady flow, and safeguarding the conditions that support disciplined execution are ongoing responsibilities. By routinely observing the process, recognizing where constraints emerge, and coordinating actions across organizational boundaries, leaders reinforce system stability and support ToC's focus on identifying and managing the true limiting factors of performance. Upholding these principles

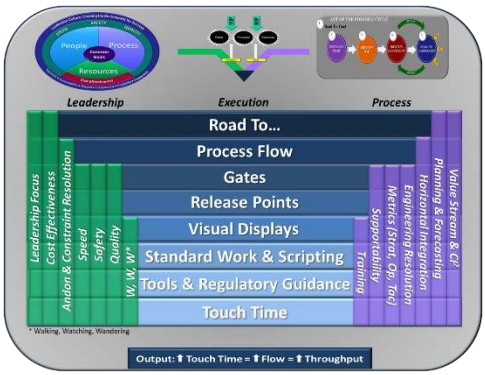
strengthens the organization's ability to deliver consistent results while establishing a solid foundation for the more advanced practices addressed in the chapters to follow.

Chapter 4

IMPLEMENTATION AND OPERATION

4.1. Introduction: Now that previous chapters have explored the necessity of focused leadership and the scientific principles of achieving optimum throughput by maximizing flow, this chapter will delve into how to execute implementation of AoP. A model, referred to as the Radiator Chart, has been developed to be a guide through the challenges of implementing AoP and ToC. It is aptly named due to its resemblance to a car radiator. While the Radiator Chart may be most helpful for a complex process, the concepts are applicable to all work, including administrative processes performed by a single person. This chapter will explain each aspect of the Radiator Chart and clarify any mystery and/or misunderstanding of how it applies to every application to achieve world class performance.

4.2. The Radiator Chart: A radiator is essential to keep a car's engine running properly. The cooling fluid within the system circulates through the radiator to be cooled. Without the radiator, the car's engine would overheat causing catastrophic failure. In much the same manner, the elements of the Radiator Chart are essential to a well-functioning process.



The Radiator Chart brings the leadership focus and the science of throughput together into a single game plan which represents the vision of how a process is set up to achieve world class status and AoP results. Within the chart, the horizontal and vertical elements intersect to represent the complexity and interdependence of its components. Focusing on select areas of the chart in isolation will not translate to success. Success depends on focus and implementation of all areas of the chart. The enterprise approach is woven into the elements throughout the Radiator Chart. The entire enterprise must align metrics and objectives of each element for the whole to be successful. Leaders need to utilize and leverage the unique capabilities encompassed within each element of the chart.

At first glance, the Radiator Chart seems complex because it is representative of the complex business of providing readiness for the Air Force. The stacked and overlapping design signifies the interdependence of the chart elements, just like each AFSC organization is dependent on one another. The length of each element signifies the ability and scope of influence each particular element has over the other chart elements**.** The Radiator Chart illustrates the standardized set of operating principles to set up processes and execute the game plan. The elements on the chart touch every part of the enterprise. The first four horizontal elements depict how the process is set up. The lower four horizontal elements provide details of how to manage the process. The vertical elements enable the process to work effectively and efficiently. Without these elements working together, the process is in danger of failing. The following is a brief discussion of each component of the model.

4.2.1. Horizontal Elements: There are eight horizontal elements which represent the blueprint for setting up and managing the Process Machine. A Process Machine is defined as the formalized flow of work broken down into segments called gates with defined entry and exit criteria known as release points. The Process Machine is guided by a Road to… Goal, underpinned by the current common goals of the organization. These are in order of strategic to tactical. Each of these will be discussed in greater detail in the following sections.

4.2.1.1. Road to… Goal: The Road to… Goal element communicates the need for a future-state goal used to set the pace for throughput and focus the enterprise in the same direction. It is the destination for accomplishing AoP results. It includes the process of communicating the goal up, down, and across the enterprise and requires stakeholder's ownership and integration of the goal into their objectives. Most importantly, it must align directly and clearly to the customer's requirements and needs. The Road to… Goal is different than common goals; however, they cannot be developed independently. The Road to… Goal is a forward-looking vision that prepares the organization for achieving the anticipated common goals of the future.

4.2.1.2. Process Flow: Process flow is a visual representation (or map) of the tasks required to complete the execution plan paced to the Road to… Goal. It defines the critical path and serves as the basis for creating standard work with repeatable, disciplined processes with predictable outcomes. The map of the process flow serves to eliminate gaps or duplication and allows users to interface and tactically manage the Process Machine to control WIP and flow. A well-structured process flow identifies predecessors, successors, and concurrent work along the critical path. A well-structured process flow also allows the identification of constraints by showing where WIP stacks up within a process and allows users to expeditiously attack and resolve their constraints or disruptions to ensure steady flow of WIP. Understanding and protecting the critical path during execution is a foundational concept throughout AoP.

KEY CONCEPT: UNDERSTANDING CRITICAL PATH

The critical path is the sequence of activities that must be completed by a specific time to meet your need date. **Time lost on the critical path is time lost for the entire system.** Leaders must urgently protect the critical path by identifying, elevating, and resolving any constraints that threaten to delay it.

4.2.1.3. Gates: The term gates refer to the practice of breaking long flowtime Process Machines into "segments" or discrete increments of work along the critical path with tangible ending points. Gates allow simplification of complex processes and projects into understandable and measurable segments of work. The use of gates creates a disciplined monitoring system with a focus on critical path urgency. CPI efforts should be tied to improving the performance of constrained gates.

4.2.1.4. Release Points: Release points within the gated process instill both the mindset and the discipline to not pass work and problems to later gates, especially as it relates to the critical path of the process flow. This includes the discipline to never release work from one gate to the next that is not supportable in respect to personnel and resources. Release points require business rules and checklists to create the culture and awareness that ensure specific actions are taken by critical points in the Process Machine. Creating a culture that uses these rules to create the urgency

necessary to elevate and resolve issues prior to the release point to protect the critical path is essential to creating the type of throughput that leads to attaining AoP performance.

4.2.1.5. Visual Displays: This is one of the elements that can answer “How do you know you are having a good day?” Visual displays are located at the point of execution (where the value is created) and provide information about the process or project. The information displayed should include measures of speed, safety, quality and cost effectiveness as well as the critical path execution of the project as applicable.

KEY CONCEPT: VISUAL MANAGEMENT

Visual management uses visual displays to actively highlight problems, guide behaviors, and drive immediate actions. To be effective, visual displays must quickly answer one simple question: **"Are we having a good day?"** To drive true transparency, these displays must be relevant, simple, accurate, and located at the point of work to bridge the gap between seeing data and knowing exactly what to do next.

4.2.1.6. Standard Work and Scripting: Creating standard work through scripting efforts allows resources to be synchronized to the needs of the product/project during execution. The focus is to create repeatable processes, which lend themselves to total transparency and are designed to eliminate constraints and enable a predictable repetitive process.

4.2.1.7. Tools & Regulatory Guidance: This element involves giving the process doer what they need. It addresses all tools required in all areas of the organization that surround and impact the process doer and the critical path. If this element is achieved properly the next element, touch time, will be positively affected.

4.2.1.8. Touch Time: This element involves keeping hands on the product/project. Kitting of materials and information needed during execution is an example of touch time enhancement efforts. Touch time improvement is about finding anything that surrounds the process doer that can positively affect their output along the critical path.

4.2.2. Green Vertical Elements: These vertical elements are focused on the leadership aspect of the enterprise and are used to set the organization up for success. Leadership sets the tone for effective issue elevation and resolution leading to the execution of efficient processes and achievement of an AoP mindset by the organization.

4.2.2.1. Leadership Focus: Leadership is responsible for building the proper environment. Leadership must continually drive their organization to understand where the problems are and determine how to resolve those problems to move the organization forward. Leadership must be comfortable in the red when setting Road to… Goals that stretch the performance of the Process Machine. Being comfortable in the red does not mean it is okay to miss customer requirements (common goals). Leaders should never be satisfied with the status quo.

4.2.2.2. Cost Effectiveness: This element involves measuring the impact of processes and output. As processes improve, output should increase without increasing cost. Savings should start to be seen in the form of time and money.

4.2.2.3. Andon & Constraint Resolution: Andon is a Japanese word meaning lantern. Within AoP, an Andon is the point at which an individual or team exhausts all options within their authority and influence to resolve an existing or impending work stoppage at a process or individual unit of work level. Andons should be quickly elevated to the next higher level with additional authority and influence to engage the enterprise and resolve the problem.

4.2.2.4. Speed, Quality, and Safety: These are the three important components of an AoP mindset. Speed – a focus to make processes effective and efficient; meeting the needs of the customer in terms of products and services. Quality – because our warfighters deserve the best. Safety – because it is the responsibility of leaders to protect the workforce. Lead with safety and quality – speed will improve.

4.2.2.5. Walking, Watching, and Wandering (W3): Participating in W3 will aid in understanding what hinders achieving maximum touch time of the doer. The term "Management by Walking Around (MBWA)" has been used through the years. This can be a component of W3 because it is a valuable method to build relationships and open communication, but W3 should be more focused on "going to Gemba" or "Gemba walks". Gemba walks are a structured, process-focused approach where leaders go to the actual place where work happens to observe processes, engage with employees, and identify inefficiencies or opportunities for improvement. The primary goal is to understand how value is created, ask insightful questions, and drive continuous improvement. Unlike MBWA**,** which is informal and focused on building relationships and morale, Gemba walks follow a purposeful plan to gather insights leading to measurable process enhancements. By regularly conducting Gemba walks, leaders develop a deeper understanding of operations, empower employees, and foster a culture of ongoing improvement. Gemba walks also provide the opportunity to observe when the doer is walking, watching and wandering because issues and constraints have diverted them from the task at hand. When choosing to "go to Gemba" remember that what the ears hear can be deceiving but what the eyes see is real (i.e., "the ears lie, the eyes do not").

4.2.3. Purple Vertical Elements: These vertical elements are focused on the processes which enable success with the help of the enterprise. Leadership will utilize these "tools" to achieve the AoP mindset throughout their organization. These elements are not intended to just improve performance but will deliver sustained and enduring resiliency.

4.2.3.1. Value Stream and CPI: Data from established gates and release points gives the information necessary to identify problem areas to focus process improvement efforts. Success should be measured against the Road to… Goal. Success is measured by results, not activities and comes from obtaining knowledge from the level closest to the process.

4.2.3.2. Planning/Forecasting: Good planning translates into good forecasts that allow all partners to strategically plan for the needs of the enterprise. Collaborative planning with all functions in the enterprise (e.g. Systems Program Office (SPO), Facility Engineers, Maintenance Planning and Production, Defense Logistics Agency (DLA) translates into better forecasts for requirements which allows a proactive approach to supportability.

4.3.3.3. Horizontal Integration: Horizontal integration speaks to the increased "synergy" possible when all members of the enterprise adopt and work toward the Road to… Goal.

4.3.3.4. Engineering Resolution: A project or process is governed by rules, regulations and instructions that provide the necessary guidance for the process doer. Situations are often

encountered that are not addressed in available guidance. When this occurs, it is essential that the required guidance is given in a timely manner to keep the project moving forward on its critical path. Engineering must also implement Process Machines to manage the variety of engineering work such as facility, equipment, process, software and other projects to provide speed and quality for the customer.

4.3.3.5. Metrics (Strat/Op/Tac): Metrics are the foundation of a data-driven organization and must be aligned from the strategic through the tactical levels. Metrics should be clear, actionable, and relate to the critical path of the project/process. However, leadership discernment is required to react to data and metrics to allow experience to drive interpretation of the data as it translates to action.

4.3.3.6. Supportability: Supportability involves proactive strategic and operational level actions to make work supportable based on findings and experience at the tactical level. Aggressive constraint identification-elevation-resolution efforts at the tactical level keep the plan executing along the critical path.

4.3.3.7. Training: Training focuses on the process doer and is linked to their tasks. It also involves training process doers to elevate problems and needs because having what is needed eliminates the push to "do what it takes."

4.3. Implementation: The Radiator Chart may seem intimidating at first glance, but the model walks users through how to define and set up a Process Machine and provides tools to facilitate management of it. The first four horizontal bars of the chart help users to set up and define the Process Machine. The following paragraphs demonstrate how to apply the horizontal bars of the Radiator Chart to any process within AFSC.

4.3.1 The Road to… Goal: For any journey to succeed, the destination must be clear. A goal cannot be achieved if it is not understood. As discussed in the Leadership Model, common goals serve as the backbone of an organization. They align all levels and partner organizations around shared objectives. These goals eliminate competing priorities, drive unified action, and define success across four key components: speed (throughput), safety, quality, and cost-effectiveness. When clearly communicated, common goals rally the organization, fostering accountability, transparency, and teamwork.

The Radiator Chart introduces the concept of Road to… Goals, which form the spirit of AoP. While common goals anchor present operations, Road to… Goals stretch the organization beyond current performance. These forward-looking objectives challenge teams to develop new skills, processes, and capabilities in anticipation of future needs. These needs may include advancements in technology, shifts in customer expectations, or evolving global security demands. Rather than vague aspirations, Road to… Goals are strategically aligned targets. They bridge today's performance with tomorrow's potential and promote continuous improvement while cultivating adaptability.

To build a future-ready organization, both common goals and Road to… Goals must be integrated into a cohesive strategy. Common goals support operational excellence and ensure alignment with the current mission. Road to… Goals, on the other hand, encourage innovation and capability

development. Together, they form a dual-track approach that balances present requirements with future growth.

In situations where the organization is underperforming, the line between these goals may become blurred. In these cases, achieving current common goals becomes the immediate stretch that prepares the organization for future progress. A structured framework is essential. Common goals should be clearly defined, measurable, and mission aligned. Road to… Goals should push organizational boundaries while remaining connected to the core purpose.

Note:

Sometimes common goals and Road to… Goals may be the same.

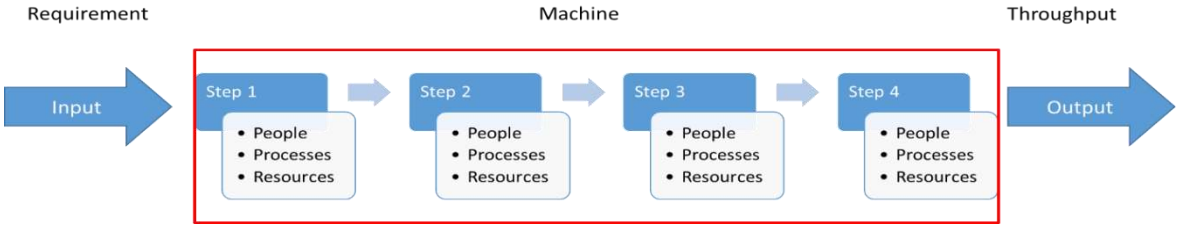
When the organization is not currently meeting its established common goals, these goals effectively become the stretch targets. This requires a focused effort to bridge the gap between current performance and desired outcomes.

The AoP management system supports balance by focusing both on current and future goals. It helps identify and prioritize Road to… Goals that address anticipated changes in global events, customer needs, and technological advancements. This roadmap approach connects short-term performance with long-term vision. It ensures meeting today's goals contributes to the ability to achieve tomorrow's breakthroughs.

Ultimately, combining common goals and Road to… Goals strengthens both agility and alignment. It ensures all levels of the organization, from leadership to the front line, operate with a shared understanding of immediate priorities and long-term aspirations. This integrated approach drives consistent performance and builds a resilient, future-ready organization that thrives in both stability and change.

4.3.2. Process Flow: Every organization produces their product or service through a system which utilizes people, processes, and resources to achieve their mission. Every process is made up of a sequence of tasks to be completed on schedule to meet the delivery date. This may sound simple, but for complex processes, the work can quickly get out of hand, and the process will get off track. Requirements enter the process and flow through in a defined path with an expected outcome. Work is performed on the product or service creating value to the customer. Good flow consists of orderly and continuous movement of the work through a series of established steps to achieve a predictable output. The graphic below represents a simple process flow. Continuous, streamlined flow of work through the process will result in maximized throughput or speed. This does not mean an organization can ignore safety, quality and cost or take shortcuts to achieve speed. Speed is achieved by improving processes, eliminating wait time, avoiding accidents, preventing quality

misses, and avoiding rework. Improving the flow through the process improves throughput and when done correctly, will also improve safety, quality and cost effectiveness.



4.3.2.1. Define the Flow: To resolve a constraint, it must first be *visible*. To be visible, the flow must be well defined and formalized. For complex processes, complete visibility can only be achieved through detailed mapping of the flow. Graphical displays should be created to allow quick communication of the flow. The following must be defined to adequately monitor the flow of work through the process.

4.3.2.1. Work Sequence: The first step of defining the flow is to determine the sequence of work to be performed. This is essential to ensure a quality product is produced and to understand the progress of the work through the process. Work requirements must be reviewed, and the process doers and enablers must be engaged, to ensure the sequence of work is logical and efficient. The work sequence should be captured in the appropriate data systems and displayed graphically, as necessary, to communicate expectations.

4.3.2.2. Critical Path: The critical path is the sequence of dependent tasks which result in the longest flowtime from start to finish. Tasks on the critical path are *dependent*, meaning one must be completed before the next can begin. Understanding the critical path of the project allows the organization to prioritize which tasks require a greater focus. Those tasks falling along the critical path directly impact the ability of the organization to complete the project on time. If any task on the critical path is delayed or underperforming, it becomes an issue to on-time delivery. Management of flow using critical path is common for complex, long duration processes with many tasks and resource requirements. Understanding and defining the critical path should be accomplished using the following steps.

4.3.2.2.1. List the Steps: As with any methodology used to manage the flow of a process, critical path begins with listing all tasks required to complete the requirement.

4.3.2.2.2. Estimate the Durations: The duration of every task should be identified and formalized

4.3.2.2.3. Identify Dependencies: All task dependencies should be identified and defined. Even if tasks are not on the critical path, determine where they will eventually impact the critical path if not completed. Dependencies should be defined by work sequence, not individual skill sets or resource requirements. For example, the fact Joe typically does two tasks on the critical path, and he cannot work both simultaneously, does not create a process dependency. Proper analysis of tasks on the critical path can help identify tasks that can be performed concurrently and moved off the critical path. This can help shorten the critical path and thereby increase throughput.

4.3.2.2.4. Determine the Milestones: Define the major phases of work, or major transition points in the flow.

4.3.2.2.5. Once all this information is collected and defined, the flow can be observed, analyzed and managed to protect the performance of the process. When an issue is determined to be a constraint to the critical path, the ToC focusing steps should be used to resolve the constraint.

4.3.3. Gates: With long flowtime processes, it is critical to break the process into smaller sections or gates. This provides increased transparency into the performance of the process, enables more timely issue identification-elevation-resolution, and ensures optimum performance of the overall process. Gates create a common language to help communicate the status of the work and issues affecting throughput to enterprise stakeholders.

Gating is the grouping of similar work within a process' overall flow. The primary purpose of the gate is to provide succinct data to pinpoint underperforming processes and clearly illustrate where to apply CPI techniques to improve performance. As such, it is imperative that gates are constructed to allow this type of visibility. Key factors to consider in the gate construct include work content, concrete (rather than abstract) boundaries, length, and total number of gates. These factors work together to construct gates which provide repeatable comparisons.

The length of a gate is an important consideration when constructing the process to guard against constructing gates that are too short or too long. Gates should be long enough to show a significant portion of the process and can encompass several handoffs between skill sets to make the gate a meaningful length. Similar work scope and content are a key determinant when constructing gates. For example, all prep work completed before a product enters a primary repair location could be grouped together, even if there are hand-offs within the gate, to make the gate a significant representation of the Process Machine.

Gates need to show the processes, but do not need to detail every sub-process or sub-task within the specific gates. It is also important to ensure the gates are not so long it becomes difficult to monitor progress or determine failure points within the gate. The ability (or inability) to consolidate gates based on work scope and content, the Process Machine critical path, and actual product physical constraints, work together to determine the number of gates required within a Process Machine.

The application of Little's Law is just as critical to the design of the individual gates as it is to the overall Process Machine design. Look at the process discussed in the section concerning Little's Law. Here the process is further broken down into gates.

Available Time (Days)	Required Output	Takt (Days)		Gate 1 (Pre-dock)	Gate 2 (Inspect Dock)	Gate 3 (Structures)	Gate 4 (System Ops)	Gate 5 (Post Dock)	TOTALS
365	64	5.7							
			WIP	2	3	10	3	5	23
			Req'd Flowtime (Days)	11.4	17.1	57	17.1	28.5	131.1

In this example, the overall flowtime is broken down into five separate gates. Defining the WIP in each gate is an iterative process depending on the physical constraints of the system and/or the amount of work to be accomplished within each gate. In some cases, the WIP will be determined by constraints such as dock space or equipment availability. In other cases, the WIP may be calculated using Little's Law based on the targeted flowtime determined through process analysis. Keep in mind that the gate flowtime for the future state Process Machine may seem unattainable relative to current performance, but it is critical to properly pro-rate the required overall WIP across the entire process. The defined WIP within the gates determines the required flowtime performance for the gate. As noted above, defining a WIP of 2 in the "Pre-Dock" gate leads to a required flowtime performance of 11.4 days (flowtime=WIP x takt or 2x5.7=11.4). Remember, the sum of the WIP in each gate cannot exceed the total WIP threshold of 23 in this example.

Little's Law calculations often lead to WIP targets that are fractions. This raises the question of how WIP is handled when it is a fraction. If the number is rounded down, the flow day requirements will decrease, and when rounded up, the flow day requirement will increase. Round up or down according to the mathematical data. In the example, gate 1 has a flowtime of 11.4. This means 40% of the time, the flowtime will be above eleven, so it makes sense to round down in this scenario. In the example, gate 5 has a flowtime of 28.5. This means 50% of the time, the flowtime will be above twenty-eight, so it makes sense to round up in this scenario.

> Note:
>
> Process Machines with long flowtime present a unique challenge and may need to be broken down into smaller challenges or sub-gates within each overall process machine gate.

While the traditional gated process has been applied to production-oriented processes, it can also be applied in administrative processes where there are multiple distinct phases of work or where there are multiple handoffs. Gating any process, whether production oriented or administrative, allows for better visibility and identification of the constraint.

Below is an example of a Process Machine in an administrative area with predefined flowtimes for each gate. Little's Law is used to determine the WIP for each gate. Observe in the graphic how flowtime and takt time are used to determine the WIP target.

The Safety Office typically processes 480 accident reports per year. Air Force guidance requires completion in 30 days. The team gave themselves a buffer of eight days and set an internal AoP Road to… Goal of 22 days. There are five gates in the investigation process. The team determined how the 22 days would be broken down in each gate based on analysis of the process. Using Little's Law the WIP was calculated for each gate based on the target flowtime goal.

Available Time (Days)	Required Output	Takt (Days)		Gate 1 (Init Inv)	Gate 2 (Classify)	Gate 3 (Inv)	Gate 4 (Report)	Gate 5 (Rel of Rep)	TOTALS
365	480	0.8							
			WIP	1.3	7.9	6.6	6.6	6.6	28.9
			Req'd Flowtime (Days)	1	6	5	5	5	22

The application of Little's Law is critical to maintaining a balanced process. This enables proper resourcing of the system to ensure resources are not overwhelmed or starved by an unbalanced process. If the organizations are not monitoring the system, failure is likely. The focus should be on the processes in each gate, not the challenges of each unit of production. Individual items should be monitored as part of the tactical management system. Performance trends of the processes within each gate should be tracked; this identifies constraints to the system and creates a tool to link CPI to the constraints. Gated processes become the basis for capturing the metrics that must be communicated "from the shop floor to the Chief's door."

In a Focus and Finish process where a single person is the doer, the individual executes the work content in one or more gates in the process. All work prior to this, such as queuing, prioritization, and full kitting, must be completed before a task is allowed to enter. Once a fully kitted task enters the gate, the person executes it with full focus, maintaining strict WIP control based on the criteria established. This design ensures clarity, protects the individual and prevents a constraint. The system around the gate is responsible for feeding supportable work, while the individual is solely responsible for finishing it efficiently, effectively, and without interruption.

4.3.4. Release Points: Release points are well-defined criteria and associated business rules governing the boundaries established for the gates. These boundaries should not be abstract but rather should be solid, uncompromisable and defined rules. Having firm release points with associated business rules is key to maintaining the most efficient flow of products and projects moving through the Process Machines. Release points are the mechanism to create the process discipline necessary to see rapid improvement and world class performance. Process discipline within the gated process instills both the mindset and the resolve to not pass incomplete work and problems to later gates. Release points require business rules to create a cultural awareness ensuring specific actions are taken at critical points in the process. Creating a culture using these rules to create the urgency necessary to elevate and resolve issues prior to the gate release point (and protect the critical path) is essential to creating the type of throughput that leads to attaining an AoP mindset.

Meeting structure should be established to support the use of release points. Leaders must dogmatically enforce release points and utilize the established meeting structure to provide the enforcement. Failure to adhere to defined release point criteria only hides issues and constraints and encourages workarounds and behaviors detrimental to Process Machine performance.

WIP Limits: The previous section presented how to use Little's Law to calculate what the WIP targets should be. Work should not be released into the Process Machine, or from one gate to the next, if it violates the WIP limits, thereby resulting in multi-tasking those resources.

Travel Work: Travel work is work that should have been completed in a prior gate but is allowed to travel to a subsequent gate. Work should not travel from one gate to the next. Efforts should be made to minimize travel work to avoid the negative impact to the Process Machine. Travel work is typically allowed to accommodate a schedule or timeline but slows down the flow in subsequent

phases of the critical path. Too much traveling work puts undo pressure on the subsequent gate and can be the primary cause of not completing the gate on time. It increases the WIP level for those downstream and leads to confusion. Travel work causes multi-tasking and hides the visibility of the constraint.

Gates should be designed with concrete release points that need to occur to complete the current gate and transition to the next gate. Abstract boundaries make consistent application of gate transition decisions difficult. Review processes must be established to control the movement of WIP from one gate to the next and must ensure the product is truly ready to transition. Movement of work to a gate that adversely impacts the critical path of the subsequent gate should not be allowed. Instead, management should utilize the situation to highlight the impacting constraint and create an enterprise call to action and create a sense of urgency.

Additionally, it is helpful if the concrete boundary is part of the Process Machine's critical path. This will alleviate transitioning to the next gate without the concrete critical path boundary requirement being met. Good gate transition habits are dependent upon how well each gate boundary is clearly defined to trigger the transition.

Release points are a critical element to successfully applying AoP to any process. It increases quality and prevents slowing the flow of work due to rework and supportability issues caused by previously scheduled work not being completed. In a Focus and Finish Process Machine, release points apply as part of the full-kitting process and in completion of work criteria to ensure quality of the product. In a DBR Process Machine the rope becomes the primary element of the release point criteria.

Important Note:

In a Focus and Finish process machine, release points will constitute part of the full-kitting process as well as criteria for completion of work to ensure quality of the product produced.

In a DBR process machine the rope becomes the primary element of the release point criteria.

4.4. Process Machine Operation and Optimization:

4.4.1. Operate and Observe: Once the Process Machine has been defined and deployed using the first four bars of the Radiator Chart, focus must shift to operating the Process Machine. The organization must manage the Process Machine and analyze the performance based on the design criteria of WIP, flow days and desired throughput. Disciplined use of AoP principles will allow the process owner to better understand the performance of the Process Machine and to identify the constraint to the performance of the Process Machine.

Process Machine performance must be monitored and analyzed for sufficient time to determine where the trends are moving and where the data indicates the constraint of the Process Machine to

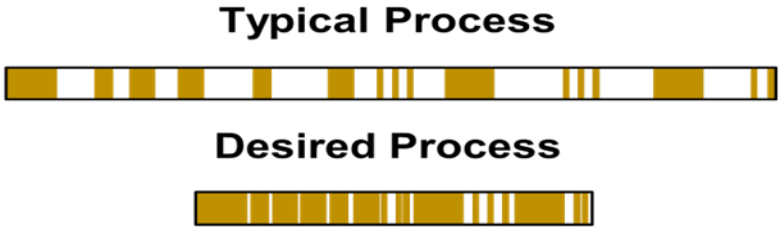
be. Operating and monitoring a gated Process Machine provides deep, actionable insight into the performance of the process by revealing exactly where and why flow is breaking down. By observing flowtime, WIP levels, and output at each gate, it will pinpoint which parts of the Process Machine are underperforming, where constraints or variability exist. It highlights bottlenecks and can show whether issues are due to lack of resources, poor kitting, rework, or upstream dependencies. Over time, trend data from each gate will predict throughput, improve planning accuracy, and tie CPI efforts directly to measurable constraints, while keeping the focus on process behavior, not just individual performance.

4.4.2. Optimize: After the Process Machine has operated for a sufficient period of time to allow the performance to be understood, the organization must turn their attention to optimizing the Process Machine. Implementing AoP is not just about printing a graphic of the Process Machine with some gate charts and talking about it once a week in a meeting. To achieve world class, there must be a focus on improving performance. Leaders must embrace the principles of the Leadership Model and embrace a sense of urgency and accountability to improve. AoP, by definition, is a relentless pursuit of achieving the “possible”.

4.4.2.1. Focus on Touch Time: The last line on the Radiator Chart is touch time. Touch time should not be viewed as a step to be implemented, but a destination to be reached. To achieve optimum throughput, doer touch time must be maximized. The doer is the only person which adds value to the product or service. How should touch time be defined? There may be a temptation to define touch time as any time that can be charged to the customer or ensuring people stay busy. However, touch time must have a much narrower interpretation. Consider it from the customer's perspective: what would the customer be willing to pay for if given the option to choose from all the doer's daily activities?

Drawing from the Joe's garage example, customers understandably object to surcharges for non-value-added activities, such as time spent locating tools or procuring parts. While these tasks are necessary for the service provider to focus on the core repair, customers perceive them as internal inefficiencies, not billable services. Similarly, clients are unwilling to pay for time a 'doer' spends searching for information or chasing details not provided in the initial request. Instead, they expect to pay for direct, productive effort: installing components, completing purchase requests, and finalizing personnel actions. Therefore, maximizing 'touch time'—the direct execution of value-producing activities—is paramount.

In the vocabulary of Lean Manufacturing, maximizing touch time is about removing waste from the doer's available time. When evaluating the doer's available time, much of the day is already accounted for by required meetings, breaks, and other necessities such as training and leave.



The remaining time must be protected to the maximum extent possible. The illustration above demonstrates how much waste can exist in a typical process. The gold-colored blocks represent the amount of value-added work, and the white space represents all the time used making up for the lack of supportability and planning or dealing with one-time interruptions. When the white space in a process is removed, the percentage of touch time to available time increases, flow time decreases, and throughput goes up. While complete waste removal from a process is impossible, the intent should be to minimize it to the greatest extent possible.

4.4.3. Find the Constraint: In the ToC focusing steps, identifying the constraint as the first step is crucial for several reasons. The constraint is the bottleneck limiting the overall performance of the system. By identifying it first, improvement efforts can be concentrated on the most critical part of the system, rather than expending resources on non-bottleneck areas. This allows for maximizing the impact of improvements, as enhancing the constraint can increase the overall throughput and performance of the system.

Constraint identification also helps avoid local optima, where optimizing a non-bottleneck area can worsen overall system performance. By focusing on the constraint, waste and inefficiency in the system can be reduced, such as excess WIP, unnecessary processing, or idle time. Additionally, ToC is based on the idea that complex systems can be improved with simple, yet effective solutions. By focusing on the constraint, simple and elegant solutions having a significant impact on the system can often be found, instead of attempting to implement complex and costly solutions that may not address the root cause of the problem.

By identifying the constraint as the first step in ToC, improvement efforts can be prioritized, and a cascade of improvements throughout the system can be created. Once the constraint is identified and addressed, new constraints will arise. By addressing them in sequence, a continuous improvement process is created. This approach ensures improvement efforts are focused, effective, and have a significant impact on overall system performance. Following this methodology unlocks the full potential of the system and achieves significant gains in productivity, efficiency, and profitability.

4.4.3.1. Operational Management Meetings/Wall Walks: Operational management is not just about keeping things moving; it is about ensuring the most critical part of the system, the constraint, is always supported, protected, and maximized. Whether in production or administrative work, every system is made up of a network of interdependent processes. But only one point at any time limits the overall performance: the constraint. Effective leaders focus their attention here, because improving the constraint is the only way to improve the system as a whole.

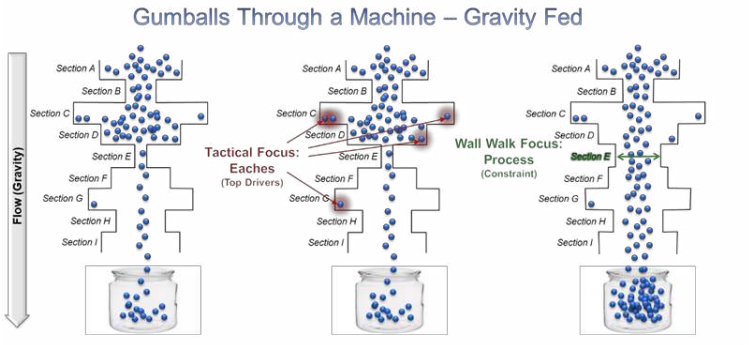
To manage operations with constraints in mind, leaders must commit to a disciplined, system-wide approach that makes constraints visible and actionable. One of the most effective tools for doing this is the Wall Walk, a recurring, process-focused review designed not just to track performance, but to reveal where flow is breaking down. Wall Walks help identify where work is piling up, where delays are occurring, and where the system is being starved, all of which are signs the constraint is being missed or mismanaged.

A Wall Walk is more than a visual review of metrics; it is a structured opportunity for the organization to surface and solve problems impeding flow. Visual displays are organized around key process gates, showing trends, triggers, and process health in real time. The most important outcome of a Wall Walk is clarity: Where is the constraint today? What is preventing it from flowing? Who owns resolving it?

Finding the constraint requires the involvement of everyone, from the front line to senior leadership. Wall Walks bring process doers, owners, and leaders together to build a shared understanding of system performance. When done well, they focus the entire organization on one question: How is the constraint being supported? Measures and metrics should be specific, data-driven, and aligned to speed, safety, quality, and cost-effectiveness, but above all, they must help expose what is limiting flow.

The power of the Wall Walk lies in the creation of ownership. When people closest to the work own the gaps, the improvements, and the actions, the organization becomes self-correcting. The role of leadership is not to solve every problem, but to build capability, guide prioritization, and remove roadblocks, especially when those roadblocks are choking the constraint. Wall Walks become a leadership platform for coaching, encouraging critical thinking, and reinforcing the focus on the true lever for improvement.

Unfortunately, many organizations tend to focus on tactical-level discussions and "top drivers" or eaches during Wall Walks, rather than truly examining the performance of the process. Top drivers refer to the specific items or issues causing the most problems or delays in the process. Use the analogy of a gumball machine to illustrate the difference between focusing on individual items (gumballs) versus examining the overall process. By focusing on the individual gumballs (top drivers), organizations may focus only on outliers and not truly understanding the performance of the process. Instead, a Wall Walk should focus on identifying the constraints in the process and developing a plan to improve the overall flow.



In the gumball machine analogy illustrated below, the constraint is section E, which limits the overall flow of gumballs. By expanding section E, the overall flow of gumballs can be increased, even if there are still some individual gumballs that get stuck. There must be dedicated time, such as a Wall Walk, to focus on the overall flow and make the overall Process Machine better. Do not allow focus on the eaches to overcome focus on Process Machine improvement through constraint identification and resolution.

In addition to Wall Walks, other operational meetings such as staff meetings should also play a crucial role in operational management. Standing meetings can provide a regular opportunity to engage in operational management, ensuring the RoF are being followed. By using the Leadership Model as a filter, leaders can address the health of the Process Machine in terms of speed, safety, quality, and cost-effectiveness. By recognizing and celebrating small successes along the way, leaders can encourage out-of-the-box thinking and create a culture of engagement. However, it is essential to avoid complacency and keep the focus on the larger goal, using Wall Walks and staff meetings as tools to drive continuous improvement and achieve operational excellence.

Ultimately, the key to successful operational management is to create a culture that values continuous improvement and employee engagement. By using tools like Wall Walks and staff meetings, leaders can empower their teams to take ownership of processes and drive improvement initiatives. By recognizing and celebrating successes, providing guidance and encouragement, and fostering a culture of critical thinking, leaders can unlock the full potential of their organization and achieve operational excellence.

4.4.3.2. Constraint Resolution Plan: The deliverable of Wall Walks and other operational management meetings should be a formal plan to resolve the constraint. Constraint resolution plans are essential because they enable and strengthen the discipline of intentionally eliminating constraints. The constraint sets the pace for the entire system, and any issue impacting it directly reduces overall throughput. Without a plan in place, problems at the constraint often lead to confusion, delays, and finger-pointing. With a defined resolution plan, the organization is prepared to respond immediately and decisively, keeping the constraint productive and protected.

A well-crafted constraint resolution plan should clearly outline how problems will be prioritized, escalated, and resolved. It defines roles, responsibilities, and response times, ensuring no time is wasted when the constraint is threatened. This alignment transforms constraint resolution from a reactive scramble into a proactive system. It enables the team to stay focused on flow, reduce downtime, and continuously improve. In essence, constraint resolution is only as effective as the plan behind it.

4.4.4. Exploit the Constraint: Once the constraint has been identified, the constraint resolution plan should become the focus of Wall Walks and operational management meetings. Experience has shown the solution is not as complex as it might seem. The ToC focusing steps indicate the first step of resolving the constraint should be exploiting it, and the best way to exploit the constraint is to apply the RoF.

4.4.4.1. Apply the Rules of Flow: Applying the RoF is critically important in exploiting the process constraint, whether in production or administrative environments, because they align the entire system around the reality that the constraint sets the pace for everything. If the system is not managed to support the constraint, throughput will be lost, lead times will increase, and chaos will ensue.

The first rule, controlling WIP, is essential because excess WIP creates clutter, hides problems, and slows everything down. By limiting WIP, the constraint is never overwhelmed, and it can

operate in a steady, predictable manner. This applies equally to the shop floor and the office, where overloaded inboxes or multitasking dilute focus and delay results.

The second rule, inducting only supportable work, ensures everything required to complete a task (the “full kit”) is available before the work begins. Starting work without everything in place leads to stop-start behavior, waste, and frustration. Ensuring full kits means the constraint never waits due to missing materials, information, or approvals. This protects its time and maximizes its throughput.

The third rule, assigning work using a synchronized plan, is about prioritizing the right work at the right time. Without this synchronization, priorities shift randomly, and firefighting takes over. The constraint ends up with the wrong work or nothing at all. A clear daily plan ensures the constraint is consistently fed and all other resources are working in sync to support it. This applies to both physical production lines and knowledge work where clear coordination between teams is vital.

Finally, the fourth rule of resolving issues quickly ensures small problems do not snowball into major disruptions. If an issue threatens the flow to or from the constraint, it must be solved quickly. In production, it might be a material shortage or equipment breakdown; in admin work, it could be a delayed decision or unclear communication. Rapid resolution keeps the constraint productive and ensures flow is not compromised.

Ultimately, these rules are not just guidelines, they are a discipline which protects the constraint, maximizes its effectiveness, and synchronizes the entire system around flow. Whether in a factory or an office, the principle is the same: exploit the constraint by creating an environment where it is never waiting, never blocked, and always adding value.

The following section discusses the application of RoF in conjunction with horizontal bars 5-7 of the Radiator Chart to help achieve bar number 8 of maximum touch time.

4.4.4.1.1. Control WIP: Controlling WIP is imperative because it is the foundation of flow, the key to exposing problems, and the most effective way to protect and exploit the constraint. When WIP is allowed to accumulate unchecked, it clogs the system, extends lead times, and creates chaos. Excess WIP hides inefficiencies, such as poor quality, unreliable processes, and unbalanced workloads, allowing the organization to operate under the illusion of productivity while critical issues go unresolved. In contrast, limiting WIP forces problems to surface quickly, enabling real-time problem-solving and creating a more stable, predictable operating environment.

This discipline does not just benefit frontline execution; it has a profound impact on supporting functions like planning and scheduling. High levels of WIP make it nearly impossible for planners to create accurate schedules or for schedulers to sequence work effectively. The variability introduced by unmanaged WIP leads to missed due dates, constant expediting, and a reactive culture. However, when WIP is controlled, planning becomes more reliable, schedules become realistic, and execution aligns with expectations. The entire organization gains confidence in its ability to commit and deliver, improving coordination between departments, strengthening trust with customers, and reducing the need for firefighting.

In both production and administrative systems, excessive WIP also causes multitasking, delay, and rework, especially in supporting roles who are forced to respond to a wider range of tasks at once, often with unclear priorities. Controlling WIP reduces this noise, allowing supporting functions to focus, prioritize, and contribute more effectively to flow. Ultimately, WIP control is not just a tactical lever, it is a strategic enabler of speed, safety, quality, and cost effectiveness, making it one of the most powerful actions an organization can take to improve performance.

4.4.4.1.2. Release Only Supportable Work: One of the most powerful levers in achieving flow and exploiting the constraint is the disciplined release of only supportable work: work fully prepared, resourced, and ready to be executed without interruption. This principle applies equally to production and administrative processes. Whether building a product on a shop floor or processing a contract in an office, releasing incomplete or unsupported work introduces delays, consumes capacity, and creates variability. It is like sending a runner onto a track without shoes, a baton, or a clear lane. If the goal is to create a predictable, high-performing system, then only work clearly defined and properly supported should be allowed into the execution stream.

4.4.4.1.2.1. Full Kitting: Full kitting is the practice of ensuring all required inputs: materials, tools, data, information, resources, and approvals are present and correct before work is released. In a production setting, this means parts, fixtures, and instructions are all in place. In administrative or knowledge-based environments, this could include access to required systems, customer inputs, policy documents, or upstream decisions. Releasing work without a full kit leads to stop-start execution, rework, and confusion. It also starves or overwhelms the constraint, hurting throughput. Full kitting enforces discipline and builds trust in the system by ensuring once work enters the flow, it can move without unnecessary disruption.

4.4.4.1.2.2. Standard Work and Scripting: After full kitting, the next layer of support is standard work and scripting; structured processes ensuring repeatability, clarity, and synchronized execution. In production, this includes defined work sequences, quality checks, and work instructions. In administrative work, it means playbooks, templates, approval workflows, and clear decision-making paths. Scripting brings alignment between people and process, ensuring work is executed the same way each time, enabling transparency, and exposing barriers to be resolved. It also makes it easier to manage constraints because variability is reduced, and performance becomes more predictable. Standard work and scripting are essential to ensure once work is released, it is executed correctly and consistently, regardless of the setting.

4.4.4.1.2.3. Tools and Regulatory Guidance: Regardless of the work environment, successful execution depends on having the right tools and clear regulatory guidance in place before work begins. In production, this includes physical tools, gauges, equipment, safety procedures and work instructions. In administrative processes, tools may be IT systems, access permissions, digital forms, or knowledge systems. Regulatory guidance may come from compliance, Human Resources, finance, or legal functions. These elements must be proactively confirmed and in place before work is released. If the doer must stop to find a tool or clarify a rule, touch time is lost, mistakes are more likely, and the flow is compromised. Ensuring tools and guidance are present and understood reduces delays and prevents costly rework, especially in regulated or cross-functional environments.

4.4.4.1.2.4. Takeaways of Releasing Only Supportable Work: The disciplined release of supportable work is a universal principle, critical to both production and administrative operations. Full kits ensure readiness, standard work ensures consistency, and tools and regulatory guidance ensure effective execution. This approach protects the constraint, improves flow, and enables teams to deliver high-quality results efficiently and predictably. Releasing only supportable work transforms reactive environments into focused, high-performing systems where work enters the process ready to win, not just ready to begin.

4.4.4.1.3. Assign Work Using a Synchronized Plan: A synchronized plan is the heartbeat of an effective operation. It tells the organization what to do, when to do it, and in what sequence to ensure resources are aligned, the constraint is protected, and flow is maintained. Without a synchronized plan, even well-prepared, fully kitted work can become chaotic, teams operate in silos, priorities shift without warning, and resources are pulled in conflicting directions. Assigning work using a synchronized plan ensures everyone is executing against the same set of priorities, grounded in what the system can actually support.

A synchronized plan does more than sequence tasks, it connects available capacity with actual demand while honoring the constraints of time, space, resources, and material. It aligns departments, functions, and teams to the same drumbeat, allowing the entire operation, whether in production or administrative settings, to function as a cohesive unit. This plan becomes the central mechanism for making informed trade-offs, responding to changes, and preventing overloading of people or processes. The key is work is not just assigned; it is committed to, with clarity about what must be completed today, what can wait, and how work flows from one stage to the next.

To bring a synchronized plan to life and keep it dynamic, organizations should have frequent tactical management meetings supported by visual displays. These tools provide the structure and transparency needed to manage real-time execution against the plan.

4.4.4.1.3.1. Daily Focus: Daily tactical management meetings are short, structured meetings held at every level of the organization from the frontline to the executive level. Many organizations refer to these tactical management meetings as tier meetings or daily standups. Frequent tactical meetings create a daily rhythm where teams check progress, align priorities, and escalate issues which threaten the plan. These meetings are essential for synchronizing cross-functional work, as they give visibility into who is doing what, where help is needed, and how today's priorities align with larger objectives. They also create a forum for reinforcing the RoF, especially the need to protect the constraint and resolve problems quickly.

4.4.4.1.3.2. Visual Displays: Visual displays are the physical or digital representation of the synchronized plan. They show today's work, current status, planned versus actual progress, WIP levels, and blockers. In production, this might be a board showing job status by station. In administrative work, it could be a kanban board showing work moving through a process. What matters is the visual display makes the invisible visible, helping teams see how their work connects to the whole system, where they are ahead or behind, and what needs attention. The best visual displays also show escalation paths, so if a deviation occurs, it is clear who is responsible for addressing it and by when.

When tactical management meetings and visual displays are used together, they create a real-time feedback loop. The team uses the plan to guide execution, the display to track progress, and the meeting to adjust as needed. This keeps the system agile, prevents drift, and maintains alignment from the constraint outward. Most importantly, it reinforces planning is not a one-time event, it is a living, daily discipline.

4.4.4.1.3.3. Takeaways of Assigning Work Using a Synchronized Plan: Assigning work using a synchronized plan is how strategy becomes execution. It is how priorities turn into performance. Without synchronization, even the best plans break down into local optimization and firefighting. By using tactical management meetings and visual displays, organizations create the visibility, discipline, and alignment needed to turn the plan into reality for every shift, every day. This rule ensures work moves with purpose, teams stay connected, and the constraint stays productive, unlocking the full power of flow.

4.4.4.1.4. Quick Issue Resolution: The fourth RoF, quick issue resolution**,** is essential because it protects the flow of work and the productivity of the system, especially the constraint. In both production and administrative processes, issues are inevitable. What matters most is how quickly and effectively they are resolved. Unresolved problems stall the flow, increase WIP, disrupt the synchronized plan, and starve downstream processes. Most critically, if the constraint is blocked by an issue, the organization loses throughput which can never be recovered. Rapid response prevents small problems from becoming systemic failures and ensures the constraint stays productive and supported.

Beyond protecting flow, quick issue resolution builds trust, engagement, and continuous improvement. When teams know their concerns will be addressed quickly, they are more likely to raise issues early, collaborate across functions, and commit to improving the process. In contrast, unresolved problems lead to disengagement, workarounds, and inefficiencies. Quick issue resolution turns obstacles into opportunities for learning and reinforces a culture of responsiveness and accountability. It is one of the key disciplines separating high-performing systems from reactive, firefighting environments.

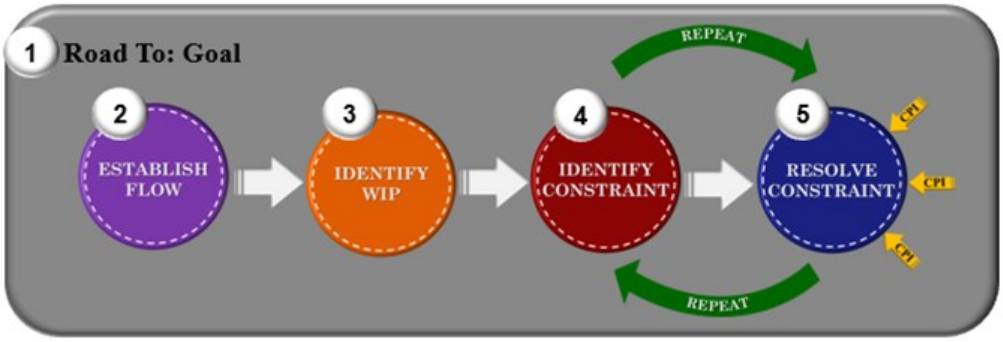
4.4.4.1.4.1. Maximize Touch Time: Quick issue resolution directly improves doer touch time by minimizing the amount of time a process doer spends waiting, searching, or working around problems. Touch time refers to the time a doer is actually adding value to the product or service. When issues are left unresolved, such as missing information, unavailable tools, unclear instructions, or bottlenecks in upstream work, the doer’s ability to engage in productive work is interrupted. Every delay pulls them away from value-adding activity and increases non-productive time.

By resolving issues quickly, interruptions are reduced, and a smoother, more predictable workflow is achieved. The doer can focus on execution rather than troubleshooting, waiting, or escalating. This not only increases efficiency and throughput, but also improves morale, reduces frustration, and reinforces trust in the system. In short, when problems are solved quickly, doers spend more time doing the work they were hired to do, which is the very definition of increasing touch time.

4.4.4.1.4.2. Takeaways of Quick Issue Resolution: The concept of quick issue resolution is crucial in maintaining the flow of work, especially at the constraint. It involves quickly and effectively resolving issues to prevent them from stalling the flow, increasing WIP, and disrupting the system. Quick issue resolution builds trust, engagement, and continuous improvement, and is essential for maximizing "touch time" (the time spent on value-adding activities). A constraint resolution plan is necessary to support quick issue resolution, particularly at the constraint, by outlining roles, responsibilities, and response times to minimize downtime and keep the system productive.

4.4.5. AoP Cycle: With the constraint identified and the RoF applied to support and exploit it, the system begins to stabilize and perform at a higher level. However, the journey to operational excellence does not end with resolution. To ensure sustained improvement and long-term reliability, it becomes essential to elevate system performance beyond immediate fixes and establish lasting solutions. This next step involves systematically elevating the constraint, addressing deeper system limitations, and expanding the capacity or capability of the Process Machine when improvement within current parameters is no longer sufficient. The following section explores how organizations can effectively elevate the constraint to unlock new levels of throughput and performance.

The AoP cycle is a repetitive cycle to be applied continuously. It incorporates the five focusing steps of ToC. These five steps (identify the constraint, exploit the constraint, subordinate everything else to the constraint, elevate the constraint, and repeat) offer a strategic roadmap for systemic improvement. The AoP cycle simplifies and operationalizes the first three steps into a repeatable discipline. Organizations can then apply the disciplines at the operational level to drive throughput gains and process stability.



Establishing flow and locating WIP leads to the "identify the constraint" step. By creating a detailed visual display of the process and showing where WIP accumulates, organizations can expose areas of underperformance, often uncovering the true system constraint. The focused execution process supports this with elements like process flow, gates, and release points, which are specifically designed to enhance visibility and control within complex systems.

Once the constraint is identified, the AoP cycle moves into the "resolve the constraint" phase, ensuring the constraint is never idle, always fully kitted, and continuously working on the highest priority items. This is achieved through rigorous application of CPI efforts to apply the RoF, particularly WIP control and synchronized planning.

Finally, by treating this cycle as a routine discipline, the AoP approach embodies the third ToC step: "subordinate everything else to the constraint." Visual management tools, leadership behaviors, and meeting rhythms described in this chapter are all structured to ensure the rest of the system aligns its actions to support the constraint. In this way, the AoP cycle does not replace the ToC focusing steps, it distills them into actionable practices to be executed at the front lines daily, forming the heartbeat of Process Machine operation and optimization within the Radiator Chart framework.

4.4.6. Operational leadership with the vertical bars: To truly unlock the potential of any operational system and achieve a state of continuous improvement, a dual focus is essential: engaged leadership and optimizing the underlying processes. The vertical bars of the Radiator Chart provide a framework for understanding and implementing these critical enablers. They represent the key elements that, when activated effectively, drive sustainable performance gains by aligning leadership behaviors with robust process design and execution.

4.4.6.1. Leadership (Green Bars): Using the green vertical bars of the Radiator Chart has a profound impact on the ability of an organization to apply the AoP principles and optimize throughput. These elements serve as the leadership enablers that activate the execution elements of the chart. While the horizontal bars depict the setup and management of the Process Machine, it is the green vertical bars which breathe life into these processes by ensuring leadership is actively engaged in driving performance, removing barriers, and fostering a culture of continuous improvement. When used effectively, these elements allow leaders to create the conditions necessary for optimizing flow and exploiting constraints, key concepts in the ToC.

For example, Andon systems and the W3 practice (walking, watching, wandering) empower leadership to see issues in real time and address them for resolution before they disrupt the flow of work. Andons signal when the critical path is at risk, allowing leaders to intervene and support resolution efforts proactively. Likewise, W3 puts leadership physically at the point of execution, enabling them to witness firsthand where constraints lie and where touch time is being lost. This aligns directly with the RoF (especially the quick resolution of issues) and creates the kind of responsive environment necessary to protect the constraint and maximize throughput. Leadership focus and cost effectiveness also play a critical role by setting stretch goals and ensuring process improvements are yielding measurable results in both time and cost.

In short, the green vertical bars translate strategic intent into day-to-day action. They reinforce leadership's responsibility to manage the system, not just the people, and drive behavior to support excellence in execution. Without these leadership elements, even the most well-designed Process Machine will fail to reach AoP levels of performance. But when leaders use them with discipline and urgency, they unlock the full potential of the system: protecting the constraint, sustaining flow, and ultimately optimizing throughput across the enterprise.

4.4.6.2. Process (Purple Bars): The purple vertical bars of the Radiator Chart represent the enterprise-level enablers which can empower organizations to achieve sustained optimization of their Process Machines. These elements are not just supplementary; they are essential to elevating performance beyond day-to-day execution and enabling a long-term, resilient approach to improvement. When actively leveraged, they align the broader system of resources, infrastructure,

data, and strategy with the operational flow, ensuring the constraint is supported not just tactically, but strategically and structurally.

For instance, value stream & CPI provide the lens and methodology to identify recurring inefficiencies and eliminate waste at the systemic level, rather than treating symptoms. Planning and forecasting anticipate future needs and allow upstream readiness, making sure material, information, and manpower are in place to support the flow before disruptions occur. Similarly, engineering resolution ensures guidance, technical decisions, and problem-solving keep pace with the Process Machine's needs, preventing unnecessary waiting on answers. By aligning metrics across all levels, organizations can maintain focus on the critical path and make data-driven decisions to support flow. These purple elements facilitate Process Machine optimization by ensuring the system is not merely reacting to issues but proactively designing against them. Without them, efforts to improve performance will stall, as tactical fixes alone cannot overcome structural limitations.

Ultimately, the purple bars create an environment in which the RoF can thrive and sustain. They ensure once the constraint is identified and exploited, the system evolves to support higher performance, without regressing or introducing new bottlenecks due to lack of alignment, forecasting, or infrastructure. In short, they are the key to turning tactical wins into strategic, long-lasting throughput gains.

4.5. Summary: The Radiator Chart, with its horizontal execution elements and vertical enabling elements, provides a comprehensive framework for achieving operational excellence. By meticulously defining the process flow, establishing gates and release points, and focusing on maximizing touch time, organizations can create a well-oiled Process Machine capable of delivering exceptional throughput. The key lies in not just building the Process Machine, but in actively managing and optimizing it through disciplined application of the RoF, continuous process improvement, and unwavering leadership focus. This holistic approach ensures the constraint is protected, flow is sustained, and the entire system works in concert to achieve AoP levels of performance.

Crucially, the vertical bars of the chart, representing both leadership behaviors and enterprise enablers, are the catalysts which bring the execution elements to life. Leadership focus, coupled with effective Andon systems and a commitment to W3, empowers leaders to identify and resolve constraints in real-time. Simultaneously, robust enterprise processes—encompassing value-stream mapping, accurate forecasting, and seamless horizontal integration—create a resilient, adaptable system capable of sustaining high performance. Without these enabling elements, even the most well-designed Process Machine will struggle to reach its full potential.

In conclusion, the Radiator Chart is not just a static model; it is a dynamic roadmap for continuous improvement. It emphasizes the interdependent relationship between leadership, process, and execution, highlighting the importance of a holistic approach to optimizing throughput. By embracing the principles embedded within AoP, organizations can cultivate a culture of operational excellence where constraints are proactively managed, flow is maximized, and performance reaches true AoP.

Chapter 5

OPERATIONAL LEADERSHIP BEHAVIORS

5.1. Introduction

To effectively implement and grow the AoP management system, leaders must bring more than just awareness, they must maintain a strong operational focus. The Radiator Chart offers a detailed structure for setting up, managing, and improving the Process Machine. But having a structure is not enough. Leaders must be present in the daily operation of the system. They must ensure teams are aligned to common goals, the right work is being released, and problems are resolved quickly. This kind of leadership is not hands-off; it requires constant attention to how work flows and where it breaks down. Without this level of engagement, even the best designed systems will underperform.

At the same time, execution cannot be separated from human behavior. People do not act based solely on plans or policies; they act based on what they believe will benefit them the most in the moment. Their decisions are shaped by what they experience, what they believe, and how certain or immediate the outcomes appear to be. Leaders must understand this reality if they want their organizations to succeed. Operational environments must be designed with human behavior in mind. That means building clarity into processes, reducing uncertainty, and reinforcing choices that lead to stronger performance and a better culture.

The following sections will explore these ideas more deeply. They will look at why people make the decisions they make, how beliefs are formed, and why accountability is often misunderstood. They will also examine the specific operational behaviors leaders must consistently practice and model to make AoP more than a set of tools. AoP must become a way of thinking and acting every day across the enterprise.

5.2. Behavioral Science

To understand why people do what they do, a simple truth must be understood: people make choices based on what they believe is in their best interest at the moment. Every day, individuals are bombarded with decisions, and they typically choose the action they think will benefit them the most, whether that benefit is avoiding pain, gaining reward, saving time, or reducing effort.

However, what a person believes to be in their best interest is typically shaped by time and certainty**,** which are two of the strongest forces in human behavior. When people see a reward coming quickly and are confident it will happen, they are far more likely to act. If a person believes with certainty that doing something will result in a positive outcome, and that outcome will happen soon, they will almost always do it. On the other hand, if the reward seems far away or uncertain, the person will often choose something else, even if the long-term benefit is greater.

This is why people sometimes make choices that might seem counterproductive. For example, someone might know that eating healthily is good for their long-term health, but they might still

eat a slice of cake. The cake provides a quick and certain pleasure. Healthy eating might not feel as rewarding in the moment, especially when results will take time to see and might feel uncertain.

In the workplace, these same principles apply. Leaders and employees often focus on short-term or tactical tasks because they give a quick sense of accomplishment. Finding a needed part or answering an urgent question brings immediate results and often results in praise. In contrast, working on improving a process might take weeks or months to show any results, and those results might feel less certain. Without visible and timely feedback, people are less likely to invest their energy into long-term improvements.

To shift this behavior, leaders must create an environment where operational management is seen as valuable and rewarding. That means setting clear expectations, giving quick feedback, and recognizing progress. When people see that working on processes will lead to real results, and they believe those results will happen in the foreseeable future and with a degree of certainty, they will choose those actions more often.

Leaders play a key role in shaping this mindset. By intentionally managing the Process Machine, they model the importance of long-term thinking. When leaders take ownership of process improvement, it shows others that it matters. As a result, employees begin to connect their personal success with the success of the organization.

In the end, people do what they believe is in their best interest. Leaders must help shape those beliefs by making the right behaviors more visible, more certain, and more rewarding. When they do, a culture of personal accountability and long-term success will follow.

5.3. Accountability: Turning Intentions into Action

Accountability is one of the six traits in the Leadership Model. But what does that really mean in a place like AFSC? AoP is best enabled in a culture where individuals and organizations consistently accept responsibility for how they execute their role. Some people are naturally driven to excel in their role and feel uncomfortable if they believe they have fallen short of expectations. They do not need anyone checking on them or pushing them. However, experience shows only a small number of people have this kind of inner drive. Most people need a structure that helps them stay focused. That is where accountability comes in.

When people hear the word "accountability," they often think about punishment. Someone messes up, and someone else hands out consequences. That might work for a short time, but it does not last. Why? Because very few enjoy punishing others. Leaders do not like doing it, and it can create tension in the workplace. Over time, leaders may avoid holding people accountable through negative reinforcement, because it makes everyone uncomfortable.

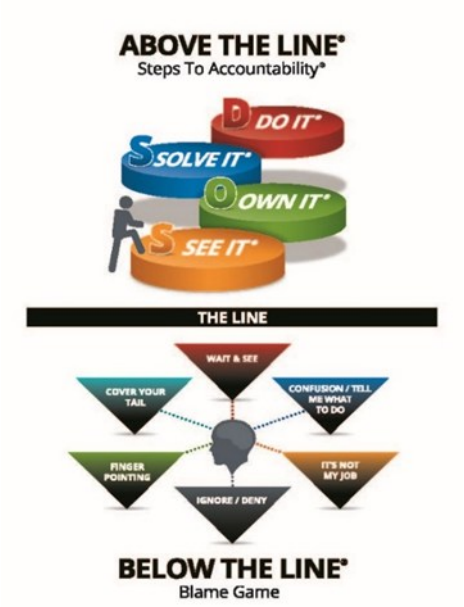
Another common issue is that people tend to push accountability off on others. People may say things like "Their supervisor did not hold them accountable" or "No one ever gets held responsible around here." Rarely do people say, "I did not do my part" or "I did not hold them accountable." But real accountability begins when someone chooses to take ownership.

True accountability works best when people want to take responsibility for their own performance in executing a job, role, or obligation. This kind of accountability is something chosen, not something forced on us. It is about caring enough to own the results, even when things do not go perfectly. When leaders take ownership of their teams, their processes, and their outcomes, they set the example for others to follow.

In the book *"Propeller: Accelerating Change by Getting Accountability Right"* (Corbridge, et.al, 2019), the authors define the **Steps of Accountability®** as choosing to rise above your situation and taking responsibility for getting the result. That means leaders do not just wait for others to fix problems. Instead, they step up, take charge, and make sure things move forward. AoP tools help leaders do this by giving them ways to find the root of problems and solve them.

The authors talk about staying ***Above the Line®***. That means choosing to be responsible instead of blaming others. When people are ***Above the Line®*** they follow the four **Steps To Accountability®:**

- ***See It®***: Individuals take a good look at the situation and try to understand what is really happening.
- ***Own It®***: Individuals take personal responsibility for their role in solving the problem, even if others are involved.
- ***Solve It®***: Individuals work to find a solution by engaging enterprise partners and employing process improvement efforts.
- ***Do It®***: Everyone takes action and follows through until the problem is fixed and the solution is sustained.



Used by Permission of Partners in Leadership

This kind of thinking is powerful. It changes people from being a victim to being part of the solution. On the other hand, people who fall ***Below the Line®*** may blame others, avoid the problem, or wait for someone to tell them what to do. They often say things like "Just tell me what you want me to do," which means they have stopped thinking for themselves and transferred the problem to someone else.

Leaders need to know when this is happening. When someone is giving up or blaming others, it is a signal that accountability is missing. Instead of ignoring it, leaders can ask questions that help the person think differently and take ownership. That helps build a stronger, more accountable culture.

One area where this really matters is in horizontal integration. That means working across teams and with mission partners to make sure everyone is aligned to the common goals. When people or suppliers fall short, it is easy to blame them. But great leaders ask if they have done everything in their power to help the situation. They work to build relationships, find solutions, and stop the problem from happening again. They also teach the people around them to do the same.

5.4. Culture: Beliefs Shape Behaviors

The outside ring of the leadership model is called Leadership Culture. This is important because culture is what gives an organization its identity. It is how teams behave, talk, solve problems, and treat each other. Good leadership creates a strong and healthy culture. The goal is not just to lead well but to build a place where everyone embodies the six leadership traits in their daily work.

Culture is powerful because it shapes how people act. But what shapes culture? The answer is beliefs. And beliefs come from experiences. What people experience every day at work (good or bad) shapes what they believe about their organization, their leaders, their coworkers and their mission.

In *Propeller* (Corbridge, et.al 2019), the authors explain this using a model known as the **Results Pyramid**®. At the bottom of the pyramid are experiences, which lead to beliefs, which lead to actions, and actions lead to results. So, to get better results, better experiences must be delivered that shape the right beliefs.

Every leader creates experiences, whether they realize it or not. Every conversation, meeting, email, or decision sends a message. That message shapes how people feel about the organization and their place in it.

Look at the six culture traits again, but this time in the form of beliefs about the enterprise:

- Teamwork: Does the organization or team work together with other mission partners to support the entire enterprise?'
- Accountability: Do teams believe everyone takes responsibility for performing their role, including leaders?
- Respect: Do teams believe they are treated fairly and with dignity?
- Transparency: Do teams believe they are told the truth and given the information they need to succeed?
- Credibility: Do teams believe their leaders can be trusted to do what they say?
- Engagement: Do teams believe their coworkers, leaders and enterprise partners care and are committed to the mission?

If the answer to any of these questions is no, then leaders need to create new experiences to change the beliefs. If the answer is yes, then leaders need to keep doing what works and find ways to make beliefs even stronger.

Culture is not built with posters or slogans. It is built one experience at a time. It takes awareness, effort, and intentionality. The way people are treated, the way problems are solved, and the way information is shared all add up to the culture we create.

Every leader has the power to shape culture. The question is whether they will choose to make things better or just let things happen. For people to believe in the mission, in their leaders, and in each other, the right experiences must be created every day.

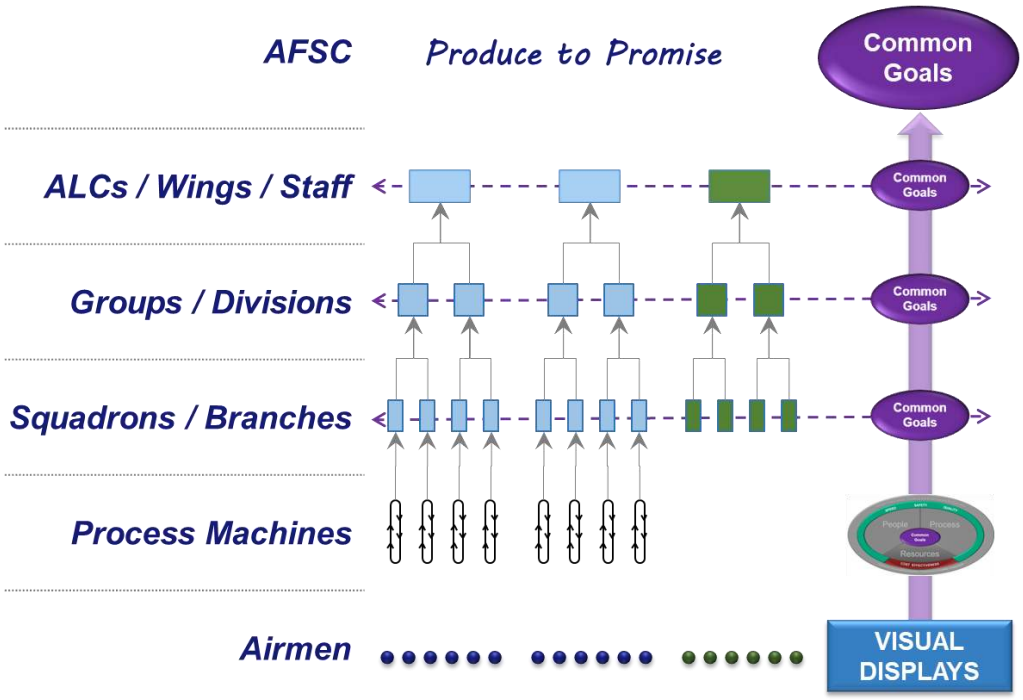
5.5. Leader Behaviors

Understanding why people do what they do and what motivates them is great in theory. But it begs the question, how is theory translated to practice? The following sections describe the core operational behaviors leaders should consistently model to turn beliefs into sustained action. While people have hundreds if not thousands of experiences every day, four leadership behaviors rise to the top that will have maximum impact in creating or improving an AoP culture. These behaviors include setting and ensuring common goals are understood and communicated, ensuring operational management meetings/Wall Walks are effective, ensuring constraint resolution plans are in place and followed, and regularly going to Gemba. These are not random acts of leadership. They are deliberate habits that reinforce **Above the Line**® thinking, strengthen accountability, and move the organization toward a more focused, unified, and high-performing culture. Each behavior creates powerful reinforcing experiences that connect people to the mission, clarify expectations, and build the confidence needed for teams to act with ownership and purpose.

5.5.1. Ensure Common Goals Are Understood and Communicated

A hallmark of effective leadership is the ability to connect the strategic with the tactical by ensuring common goals are developed, understood, and actively pursued at every level of the organization. Leaders play a crucial role in setting this tone. When they demonstrate the behavior of aligning goals from the top down, from enterprise objectives all the way to individual daily performance, they not only provide clarity but also create unity of effort. It begins with commander guidance at the center level, which lays out expectations for performance, particularly the importance of producing to promise. From that foundation, leaders must work horizontally with enterprise partners to establish shared goals across organizational boundaries. These become the framework that supports everything else.

Once enterprise-level goals are defined, leaders must make sure they are communicated down with care and purpose. Unit-level goals, whether at the complex, wing, group, or flight, must be thoughtfully developed in direct support of the broader enterprise objectives. This does not happen automatically. It requires leaders to engage in deliberate conversations, translating strategic priorities into specific, actionable goals for their teams. Without this kind of leadership behavior, organizations risk becoming fragmented, where each unit works hard but not in harmony. Leaders who commit to shaping and communicating these goals build teams that understand where they are going and why it matters. The graphic below illustrates the levels at which common goals should be understood.



At the Process Machine level, where the work takes place, goals need to be even more precise. Leaders must define the expected level of performance required for mission success. This is not simply about tracking metrics. It is about creating a shared sense of purpose among those who carry out the daily work. Leaders who take time to connect these machine-level goals to higher mission goals reinforce both accountability and clarity. They help individuals understand their efforts are not isolated but are vital links in achieving broader success.

At the doer level, leadership behavior becomes personal. Effective leaders find ways to make the mission relevant to each individual. They help workers answer the question, "Am I having a good day?" not in terms of personal comfort, but in terms of contribution to common goals. When leaders consistently link daily tasks and performance to larger objectives, they build a culture of ownership, motivation, and pride in the work.

Developing and communicating common goals is not a one-time task. It is an ongoing leadership behavior that must be demonstrated consistently. It involves translating vision into action, fostering alignment, and modeling commitment to the mission. When leaders take responsibility

for connecting each level of the organization to common goals, they create clarity, strengthen accountability, and lay the foundation for long-term success. It is not only about setting direction but about building a shared sense of purpose that brings unity across the entire organization.

5.5.2. Ensure Operational Management Meetings/Wall Walks Are Effective

In any high-performing organization, leaders must consistently demonstrate the behavior of making operational management meetings, especially Wall Walks, a focused and purposeful part of their leadership responsibilities. These meetings are not formal briefings, casual reviews, or passive updates. As discussed in chapter four, Wall Walks are structured working sessions designed to drive process improvement where it matters most, in the day-to-day operations of the Process Machine and specifically in the constraint. When leaders prioritize and protect these meetings, they are making a statement that improvement is central to the mission and not an afterthought. It becomes clear these meetings are not optional activities, but essential tools for organizational success.

Wall walks serve a specific function that is different from tactical management meetings. Tactical management meetings are important for managing immediate tasks and addressing individual work items, but Wall Walks are designed to focus on system-level performance. They look at trends, not isolated issues. Leaders who understand this distinction help teams evaluate whether the process is working, not just whether the work is getting done. These meetings must be data-driven, supported by visual displays, and managed by a clearly identified process owner. Leaders who create this kind of structure provide clarity and keep the team aligned with the organization's common goals.

It is equally important for leaders to ensure Wall Walks include the right people and the right focus. The process owner must be empowered to run the meeting, and the focus must remain on the performance of the process over time. These are not status briefings. They are working sessions. Leaders must resist the urge to take over or change the intent of the meeting. When leaders model respect for the structure and intent of operational management meetings, they reinforce the idea that improvement is everyone's responsibility, and the process deserves our full attention.

Action is what gives these meetings their value. Leaders must ensure every Wall Walk results in clear assignments, defined follow-up tasks, and real accountability. This is where leadership behavior makes the biggest difference. When leaders follow through, track progress, and keep the momentum going, they show these meetings are not just for discussion but for making measurable improvements. It builds trust and reinforces that time spent in the meeting translates directly into results.

When leaders regularly participate in, protect, and model the right behaviors in operational management meetings, they help embed a culture of continuous improvement. They demonstrate process performance is not just a technical concern but a leadership priority. In doing so, they move the organization from reactive firefighting to proactive, intentional improvement. This kind of leadership brings lasting value by ensuring every part of the operation is aligned with the mission and improving over time.

5.5.3. Ensure Constraint Resolution Plans Are In Place and Followed

One of the most critical behaviors a leader can exhibit in a flow-based organization is ensuring constraint resolution plans are developed and actively followed. Constraints are the points in the Process Machine that limit the entire system's output, and identifying the constrained gate is only the beginning. Effective leaders dig deeper to discover the true constraint within the gate—where the actual bottleneck lies—and then focus resources and attention there. This is not a one-time event but a continuous leadership responsibility that reinforces discipline, alignment, and accountability.

A well-constructed constraint resolution plan should naturally flow out of operational management meetings, where data reveals performance trends and barriers to flow. From these insights, leaders must help drive the creation of an event-driven constraint resolution plan. These plans must include clearly defined action items, responsible points of contact, due dates, and a calculated expected return on investment. This structure reinforces accountability by tying improvement efforts directly to measurable outcomes. Leaders who model this behavior create a culture where performance issues are addressed quickly and deliberately—not postponed or passed around.

The RoF provide the most practical and effective method of exploiting the constraint once it is identified. First, leaders must control WIP to prevent overloading the constraint, ensuring the limited capacity is not wasted on excess or premature work. Just as important, they must induct only supportable work, meaning work that is fully kitted and ready for efficient execution, rather than work that requires rework or stops midstream. In addition, daily synchronized plans must be used to assign work to the constraint in a way that maximizes flow without interruption. Leaders must also create the environment and mechanisms to resolve issues quickly, protecting the constraint from delays and disruptions. When leaders ensure these rules are being applied consistently, they demonstrate their commitment to enabling the constraint to operate at its highest potential.

Constraint resolution plans should leverage CPI tools to provide the structured methodology to not only manage but also gradually elevate the performance of the constraint. While the RoF ensure the constraint is protected and optimized in the short term, CPI equips leaders and teams with the tools to identify root causes, eliminate waste, and implement sustainable solutions that expand capacity over time. By applying disciplined problem-solving methods, leveraging data, and engaging employees closest to the work, CPI turns constraint management from a reactive effort into a proactive cycle of learning and improvement. This balance between stabilizing today's demands and systematically improving tomorrow's capacity ensures that the organization continually strengthens its ability to meet the mission.

KEY CONCEPT: CONSTRAINT RESOLUTION

A system is only as strong as its weakest link. We must build a culture focused daily on identifying and urgently eliminating constraints. Apply the five focusing steps of ToC: **1) Identify, 2) Exploit, 3) Subordinate, 4) Expand, 5) Repeat**. When exploiting, use the Rules of Flow: **1) Control WIP, 2) Release Only Supportable Work, 3) Assign Work Using a Synchronized Plan, 4) Quick Issue Resolution**. Use urgency tools like Andons to identify and resolve issues before they impact the critical path.

Ultimately, constraint resolution is about focus and discipline. Leaders who prioritize these traits model **Above the Line®** thinking by taking ownership, seeking solutions, and remaining committed to systemic improvement. They resist the temptation to chase after symptoms and instead help their teams concentrate on the leverage point that will move the whole system forward. This kind of leadership transforms frustration into flow, confusion into clarity, and scattered activity into purposeful progress.

5.5.4. Go to Gemba

The behavior of consistently going to Gemba is one of the most powerful ways a leader can demonstrate commitment to accountability, continuous improvement, and servant leadership. Gemba, meaning "the actual place," is where the work truly happens, whether that is on a production floor, in a logistics center, an administrative office, or even in a digital workspace like Microsoft Teams. When leaders make the effort to go and see for themselves, they step into the reality of their team's daily experiences. This practice not only closes the gap between assumption and truth, but it also reinforces the message that leadership is rooted in connection, understanding, and presence.

When leaders go to Gemba, they create meaningful experiences, both for themselves and for their teams. These experiences drive beliefs, and those beliefs shape behavior. Observing the work firsthand reinforces belief in the importance of the process, belief in the people doing the work, and belief in the accountability model. It is in these moments that **Above the Line®** thinking comes to life. Leaders should model curiosity instead of judgment, seek clarity rather than control, and look to support rather than blame. This creates a culture where it is safe to be honest about problems and improvement is seen as a shared responsibility.

It is important to understand that going to Gemba is not about fixing things on the spot or assigning blame. It is about seeing and listening. Leaders who go to Gemba with humility and purpose create space for employees to speak honestly about their challenges and ideas. "Ears lie, eyes do not" captures the essence of this principle. Data, reports, and briefings are valuable, but they cannot replace the truth that comes from seeing the people and process at work. These interactions, when

handled well, become reinforcing experiences that build trust and move others **Above the Line®** where ownership, learning, and constructive action take place.

To be fully effective, Gemba walks must be both planned and spontaneous. Some visits should be scheduled and protected on the calendar as a regular leadership routine, sending the message that the work and the people doing it are worth the time. Other visits should be unplanned and driven by curiosity, perhaps sparked by something mentioned in a meeting or noticed in performance trends. Whether planned or impromptu, the key is consistency. Leaders who show up regularly demonstrate they care deeply about the mission and the people who make it happen.

In the end, leaders who go to Gemba with intentionality create a culture where accountability is visible, improvement is normal, and people feel valued. They model the behaviors of **Above the Line®** thinking and, by doing so, lift others with them. These visits are not just about solving problems, they are about shaping beliefs and aligning hearts and minds to a common purpose. It is in these moments that the culture can be built and sustained, one conversation and observation at a time.

KEY CONCEPT: GEMBA WALKS

Walking, Watching, and Wandering (W3) is how leaders actively engage to answer the question, "Are people on task?" You cannot solve problems from a desk. You must solve them where they live, through direct observation on the floor, to eliminate waste, improve touch time, and cultivate a "minutes matter" mentality.

5.6. Summary

Effective operational leadership goes beyond mere awareness, it demands a consistent, hands-on focus to ensure processes perform optimally. This requires understanding human behavior. It requires understanding that people make choices based on perceived immediate and certain benefits. Therefore, leaders must intentionally design environments and experiences that make desired behaviors such as process improvement and long-term thinking more visible, certain, and rewarding. True accountability, in this context, is not about punishment but about chosen ownership, exemplified by the **Above the Line®** mindset of seeing, owning, solving, and doing. Ultimately, organizational culture, which shapes behavior, is built on beliefs formed by daily experiences, making leaders crucial architects of the employee experience.

To translate this theory into practice, leaders must consistently model four key operational behaviors. Firstly, they must establish and align common goals from the strategic enterprise level down to individual daily tasks, ensuring unity of effort. Secondly, they need to prioritize and conduct focused operational management meetings (wall walks), which are structured, data-driven sessions aimed at system-level process improvement, not just status updates. Thirdly, leaders are responsible for ensuring constraint resolution by actively identifying bottlenecks, developing event-driven plans, and applying the RoF to maximize system output. Finally, effective leaders consistently go to Gemba – the actual place where work happens – to observe, listen, and build

trust. This practice bridges the gap between assumptions and reality, reinforcing accountability and fostering a culture of continuous improvement by showing genuine commitment and building stronger relationships.

By consistently practicing these behaviors, leaders create powerful reinforcing experiences that shape positive beliefs, strengthen accountability, and drive the organization towards a high-performing culture.

Chapter 6

SUSTAINMENT AND THE MATURITY MATRIX

6.1. The Importance of Sustainment

The pursuit of excellence requires more than temporary improvements or bursts of enthusiasm. It demands steadfast commitment to sustaining the hard work already accomplished. Sustaining AoP is fundamentally about holding the gains won through effort and ensuring they become the new normal, rather than fading away with the next change of leadership or shifting priorities.

Excellence is the reality of being all we can be, both as individuals and as an organization. However, without sustainment, even the most innovative improvements risk slipping away. A new initiative might launch with posters on the walls and a flurry of activity, only to vanish silently, leaving only traces of what might have been. Such patterns are all too common in organizations worldwide.

Sustainment has many enemies: lack of standard processes, shifting priorities, leadership turnover, or simple human nature to retreat to the comfort zone of "the way we've always done it." Without standard work, there is no stable baseline. Each new leader might bring a different approach, and the processes evolve in ways that become disconnected from the Road to… Goals.

Change itself is not the enemy. In fact, AoP is built upon the belief that we should be dissatisfied with the present and seek to reach beyond our current capabilities. Yet for change to be meaningful, it must build upon stable, standardized foundations. Otherwise, we find ourselves in a state of constant churn, measuring progress against a moving target and eroding both morale and trust in leadership.

Leader standard work is one of the most critical tools in ensuring sustainment. Leaders cannot simply implement new processes and walk away, assuming the workforce will carry on as intended. Implementation is merely the beginning; sustainment is the true measure of success. Just as an army would not take a hill in battle only to leave it unguarded for the enemy to reclaim, leaders must hold the gains achieved through new processes and improvements.

Ultimately, sustainment protects the investment of time, effort, and resources that went into achieving progress. It ensures the organization does not continually find itself retaking the same ground, exhausting its people and hindering its forward momentum.

6.2. Using the Maturity Matrix to Guide Organizational Growth

Sustainment establishes stability, but stability alone does not equal excellence. Once an organization can sustain its processes reliably, it is ready to advance toward higher levels of performance. This journey from stability to world-class operations is measured and guided by the Maturity Matrix.

The Maturity Matrix is a powerful tool designed to help leaders assess how deeply the principles of AoP have taken root within their organizations and broadly across the enterprise. It provides a

structured, transparent method of measuring progress along the continuum from set-up to institutionalization and finally to achieving world-class status.

The Maturity Matrix evaluates performance across each of the eight horizontal elements of the Radiator Chart, establishing a common yardstick from the enterprise level down to the process doer's touch time. Each element is scored from Level 1 (set-up) through Level 5 (world-class), with clear, progressively more demanding criteria defining each stage. Below is a sample of the Road to… criteria.

	SET-UP		INSTITUTIONALIZATION		WORLD CLASS
	1	2	3	4	5
ROAD TO …	Goal created based on AFSC/CC Priorities, current AFMC, AFSC or Complex Strategic Plans, future customer requirements, existing capacity / capability gaps	Goal communicated to Key Stakeholders and Leaders vertically down within the organization to the Flight/Branch level	Enterprise Value Stream Mapping (EVSM) with Mission Partners has been completed; mission partners accept the Road To goal and path Goal communicated vertically up to all levels within Complex / Directorate / Wing organization WIP Reduction, CPI, and constraint resolution process are operating and linked directly to Road To Goals	Goal is communicated horizontally across AFSC enterprise, supporting external organizations across the enterprise Stakeholder metrics and leadership objectives aligned to Complex / Directorate / Wing metrics and objectives Significant Progress on Road To Goal with incremental goals achieved Targets have been adjusted to achieve stretch goals, second pass EVSM with mission pass completed; evidence of time/cost reduction impacts	ENTERPRISE ALL-IN! Goal is adopted and worked toward by stakeholders; up, down and across Complex / Directorate / Wing and Stakeholder leadership integrating and influencing strategic objectives and initiatives across all parts of the Sustainment and Life Cycle enterprise based on Road To… and cost effectiveness goals

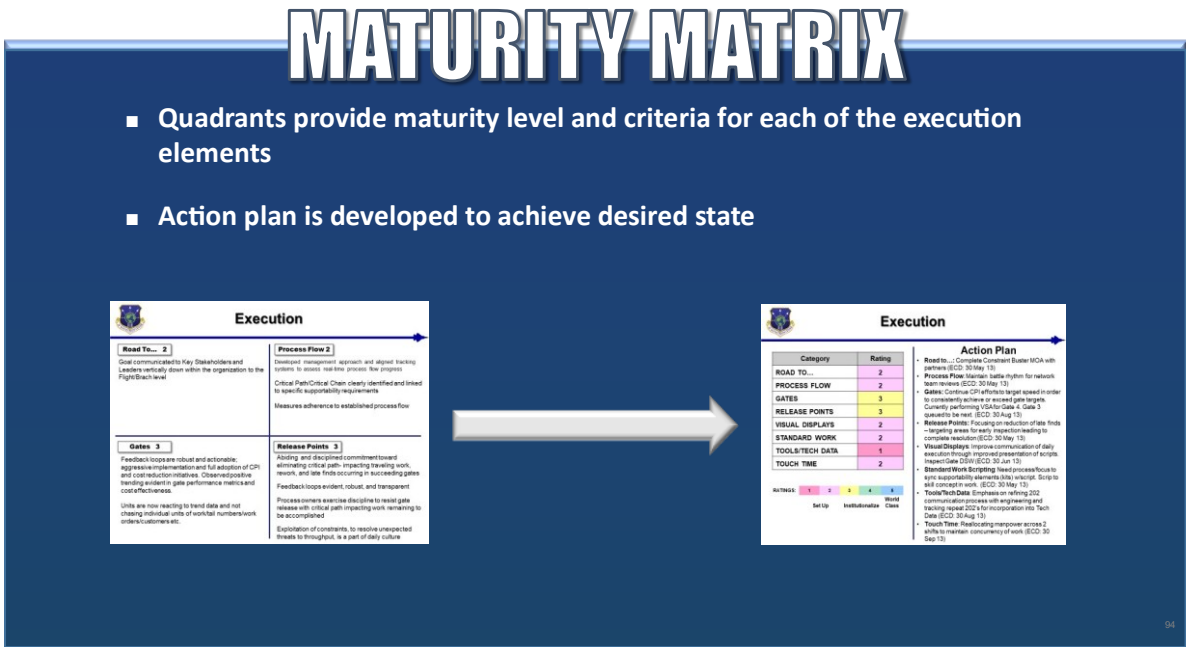
**The Complete Maturity Matix can be found on the AoP SharePoint Site.

Leaders must recognize that moving beyond Levels 1 or 2 is not merely a matter of internal effort. Reaching Level 3 and beyond requires enterprise-wide alignment, cross-functional collaboration, and even support from external stakeholders. Progression through the Maturity Matrix is intentional and demanding, designed to push organizations beyond silos and toward shared excellence.

Assessing maturity is not a box-checking exercise. The criteria may appear subjective at times, but leaders are expected to provide evidence of their assessments. They must be able to answer three critical questions:

1. What is your currently assessed maturity rating?
2. What evidence supports this assessment?

3. What actions will be taken to advance to the next level or desired state?



A maturity score is far more than a number—it reflects the state of the organization's Process Machines. It captures how robustly the organization is managing the fundamental principles of AoP, how stable its processes are, and how well it is positioned to evolve toward world-class status.

Honest use of the Maturity Matrix drives transparency, accountability, and alignment across the enterprise. It helps ensure progress is not based on perception or superficial results but on solid evidence and consistent practices. Leaders at every level should study the matrix carefully and use it as a guide not only to evaluate their current standing but to chart the path forward. Inaccurate assessments not only misrepresent maturity, but they also undermine the use of the Maturity Matrix as a ladder to greater maturity. An inflated maturity score will cause an organization to skip over necessary rungs on the ladder (criteria on the Maturity Matrix) that have not been attained and take on next steps for which it is not yet ready.

By coupling the discipline of sustainment with the insight provided by the Maturity Matrix, AFSC—and any organization striving for excellence—can steadily transform from implementing AoP principles to embodying them as the organizational way of life.

6.3. Putting It All Together

Sustainment and maturity are inseparable. Sustainment protects the progress made through AoP, while maturity drives the organization closer to world-class performance. Tools such as standard work and the Maturity Matrix provide a solid foundation, but it is leadership commitment that brings these concepts to life.

Leaders who remain vigilant in sustaining gains and continuously strive for the next level of excellence ensure AoP is not just a program—but the way their organization operates every day.

Too often, organizations stop implementing action items, believing implementation alone is enough. True sustainment happens only when leaders and teams stay the course through sustainment. It's the commitment to maintain and improve that protects the organization's progress and advances it toward higher maturity levels.

Appendix A – RECOVERING A PROCESS MACHINE

One or two ill-informed decisions or deviations from AoP principles can quickly derail a Process Machine. However, recovery is not nearly as quick. It takes a methodical and intentional approach focusing on RoF and specific constraint resolution plans to recover. A Process Machine owner must engage enterprise partners and leverage RoF to maximize supportability and touch time. Process teams will never fully recover a Process Machine or improve touch time without adhering to RoF first. Leaders at every level must recognize if something is off track and take swift corrective action on Process Machines within their organizations.

A.1. After a process owner determines the machine is set up (or gated) properly to allow smooth flow from induction to output, and there is a good understanding and agreement of the common goal, the owner must evaluate whether process stakeholders are adhering to RoF by looking at the following steps:

A.1.1. Monitor and control WIP within the machine (to include WIP within each gate). Excessive WIP within any one gate or the overall machine leads to poor touch time, spreading manpower and resources too thin, and supportability issues from your enterprise partners.

A.1.2. Ensure only supportable work is released to process doers. Determine whether all resources, materials, parts, technical data, process orders, instructions, tools, equipment, etc. are available to successfully accomplish the job. Ensure kitting is fully utilized across the process. Confirm the team is looking forward two to five days to ensure upcoming work will be fully supportable at the appropriate time.

A.1.3. If the task or job is fully supportable, check that work is being assigned using a synchronized plan such as daily stand-up meetings, tier meetings, production/support networks, or fixer release points between gates. The bottom-line is work must be assigned based on a daily/weekly scripted work plan, battle rhythm, and priorities.

A.1.4. Lastly, validate the team is practicing quick issue resolution. Never let issues impacting process flow or touch time linger. Pull together the right enterprise partners to solve these issues at the right level with the appropriate sense of urgency. Process owners must stay above the line… see it, own it, solve it, and do it!

A.2. Leaders must recognize they own the recovery effort…and must drive improvements. This requires leaders to practice the three Ws of walking, watching, and wandering in the area where the work is being performed and ensure the right amount of manpower is assigned and present each day to accomplish the work. It is important to focus on the constraint gate first. However, focusing on RoF will improve touch time and throughput in all areas.

After focusing on and truly implementing RoF, administer the five ToC focusing steps to eliminate or at least mitigate the true constraint within the Process Machine.

A.2.1. Proper visual management (databases, production control boards, dashboards…etc.) will identify the constraint gate. However, additional analysis will reveal the specific constraint causing the issues.

A.2.2. Once the true constraint is identified, exploit or optimize that constraint by getting everything out of it using the current resources, manpower, and materials available.

A.2.3. After exploiting the constraint, subordinate or collaborate to control the constraint by matching the machine pace to the pace of the constraint gate so resources, manpower, and materials are not wasted. Place focus on protecting and resolving the constraint.

A.2.4. If after subordinating everything to the constraint gate's capability it is still not resolved, expand, elevate, or upgrade the constraint through CPI efforts by hiring additional manpower, cross-training personnel, or purchasing additional or new tools, equipment, software, facilities, etc.

A.2.5. Having eliminated or mitigated this constraint, start over and look for the next constraint in the machine.

A.3. The playbook to recovering struggling or broken machines is simple… but, not easy! It takes dedicated leaders willing to step up and own recovery process by validating adherence to RoF as outlined in this handbook… all while using the ToC five focusing steps to eliminate or mitigate the overall constraint in the Process Machine.

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

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**Additional references and reading material can be found on the AoP SharePoint site.

Prescribed Forms

AF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

AFI – Air Force Instruction
AFPD – Air Force Policy Directive
AFRIMS – Air Force Records Information Management System
AFSC – Air Force Sustainment Center
AFSCI – Air Force Sustainment Center Instruction
AoP – Art of the Possible
CCPM – Critical Chain Project Management
CPI – Continuous Process Improvement
DBR – Drum Buffer Rope
IAW – In Accordance With
JP – Joint Publication
OPR – Office of Primary Responsibility
RDS – Records Disposition Schedule
RoF – Rules of Flow
ToC – Theory of Constraints
W3 – Walking, Watching, and Wandering
WIP – Work In Process

Glossary

This glossary is intended as an explanation of terms that may be new or uncommon.

Andon – Andon is a Japanese word meaning lantern. Within AoP, an Andon is the point at which an individual or team exhausts all options within their authority and influence to resolve an existing or impending work stoppage at a process or individual unit of work level. Andons should be quickly elevated to the next higher level with additional authority and influence to engage the enterprise and resolve the problem.

Art of the Possible (AoP) – A constraints-based management system designed to create an environment for success by creating a culture of problem-solvers, defining processes (aka machines), eliminating constraints, and continuously improving. It is the framework for how the AFSC conducts business and how we strive to achieve world class results in warfighter support.

Comfortable in Red – Refers to the willingness to set aggressive targets with the understanding the metrics will show as "red" until process throughput efficiencies improve.

Common Goals – Goals that should be understood and shared across the enterprise. Common goals should address speed, safety, quality and cost effectiveness.

Constraint – Systemic machine or process bottlenecks that hinder throughput.

Constraint Gate – The gate with the lowest throughput.

Continuous Process Improvement (CPI) – The focus of continuously improving the flow of work through the organization using a combination of tool sets such as: Lean, 6-Sigma, and ToC.

Critical Path – A sequence of activities in a project plan which must be completed by a specific time for the project to be completed on its need date. The AFSC adaptation of this term refers to the linkage of critical elements in a process or project that keeps the work realistically moving forward toward completion.

Drum Buffer Rope (DBR) – A schedule methodology that controls the release of work into the system. It is a pull system in the sense that when a job is completed by the constraint resource, it sends a pull signal to trigger the release of a new job into the system.

Flowtime – The average time that a unit stays in a production machine.

Kanban – A visual system for managing and improving work by visualizing the workflow and limiting work in progress.

Maturity Matrix – AFSC method of measuring organizational maturity regarding the adoption of principles found in the "Execution" elements of the AFSC Radiator Chart.

Process Machine – Refers to the science of the process and implies that any process can be gated in order to measure throughput and focus process improvement activities.

Queue – Work awaiting induction to a process. Also, a WIP control tool in a gated monitoring system.

Quick Issue Resolution – The rapid identification and resolution of issues that affect the movement of individual items through the Process Machine.

Radiator Chart – Model depicting the fundamental components of the AoP methodology.

Road to… Goal – Reflects the throughput-pace required for both the interest of the customer and the organization. The throughput goal that sets the pace of the process.

Standard Work – A detailed, documented and sometimes visual system by which team members follow a series of predefined process steps. It is how work is accomplished and is defined by technical data, process orders, regulations, instructions, or approved checklists.

Tactical Management – An established frequent review of WIP flowing through the Process Machine. It focuses on the individual items of WIP flowing through the Process Machine rather than the Process Machine performance at the operational level.

Takt Time – The rate of customer demand, how often a single unit must be produced from a machine (takt is a German word for rhythm or meter).

Theory of Constraints (ToC) – A management philosophy that identifies and manages the most significant limiting factor, or constraint, in a process or system to improve overall performance**.**

Throughput – The required output of a production machine expressed in units per time.

Visual Management – The use of simple visual indicators to help people determine immediately whether they are working inside the standards or deviating from it, this must be done at the place where the work is done.

Wall Walk (Operational Management) – A recurring process-focused review to understand Process Machine performance, to identify constraints, and to coordinate constraint resolution.